

Perioperative Medicine

Dr. Ann Marie McKenna

Friday, November 28, 2025



List of Short Forms Used in this Talk

DASI	Duke Activity Status Index https://www.mdcalc.com/duke-activity-status-index-dasi
CPET	CardioPulmonary Exercise Testing
CCS	Canadian Cardiovascular Society
METS	Measurement of Exercise Tolerance Study (or metabolic equivalents – depends on context!)
RCRI	Revised Cardiac Risk Index https://www.mdcalc.com/revised-cardiac-risk-index-pre-operative-risk
POBA	Plain Old Balloon Angioplasty (No Stent)
BMS	Bare Metal Stent
DES	Drug-Eluting Stent
LMWH	Low Molecular Weight Heparin
LDUH	Low Dose Unfractionated Heparin

The Pre-op Assessment

The Checklist:

- ☐ Cardiac risk assessment
 - ☐ + Don't forget endocarditis prophylaxis
- ☐ Delirium Risk and Frailty
- ☐ Manage Medications
 - ☐ BB, RAASi, statins, Antiplatelets
 - ☐ Anticoagulants
- ☐ Heme considerations:
 - ☐ VTE prophylaxis
 - ☐ Anemia optimization
 - ☐ Coagulopathies (liver, ITP, VWD etc.)
- ☐ Metabolic: Diabetes, stress steroids
- ☐ Respiratory Risk Stratification



Focused Physical Exam:

Gait speed, assess walk into your exam room

CNS: Delirium Risk - Executive function - *Check pills and bottles*

Cardiopulmonary exam : AS, HCM, Pulmonary HTN

Peripheral vascular exam
(+Diabetic foot pre TKA/THA)

Pre-op Case

82yF living in apartment alone, scheduled for **R hip arthroplasty in 10 days**. She has a history of atrial fibrillation, HFnrEF (LVEF 50%), CKD (eGFR 45).

She is now confined to apartment due to severe hip pain. She enjoys card games and plays online bridge with her grandchildren. She manages her meds and finances but has help for meals (meals on wheels), shopping, housekeeping, and PSW to support bathing.

Meds: apixaban 2.5mg po BID, dapagliflozin 10mg daily, furosemide 40mg daily, metoprolol 25 mg po BID, candesartan 16mg daily.

Exam: BP 132/78, HR 80-90 irregularly irregular. 98% room air. Wt: 59 kg.

Systolic ejection murmur crescendo-decrescendo, Normal s1, s2 with radiation to clavicle.

Chest clear, no peripheral edema, JVP not elevated.

Struggles to stand and walk 10m in clinic due to hip pain.

Hgb 119, ferritin 40 / Tsat 15%, plt 280

Cr 87 (CrCl 41)



Key Guidelines: CCS and ACC/AHA



Canadian Journal of Cardiology ■ (2016) 1–16

Society Guidelines

Canadian Cardiovascular Society Guidelines on Perioperative Cardiac Risk Assessment and Management for Patients Who Undergo Noncardiac Surgery

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Circulation

CLINICAL PRACTICE GUIDELINES

2024 AHA/ACC/ACS/ASNC/HRS/SCA/SCCT/SCMR/SVM Guideline for Perioperative Cardiovascular Management for Noncardiac Surgery: A Report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines

Developed in Collaboration With and Endorsed by the American College of Surgeons, American Society of Nuclear Cardiology, Heart Rhythm Society, Society of Cardiovascular Anesthesiologists, Society of Cardiovascular Computed Tomography, Society for Cardiovascular Magnetic Resonance, and the Society for Vascular Medicine



Great guideline addressing many aspects of perioperative care.
For periop Cardiac Risk stratification + antiplatelets = stick to CCS

Cardiac Risk Assessment

Evaluate Risk

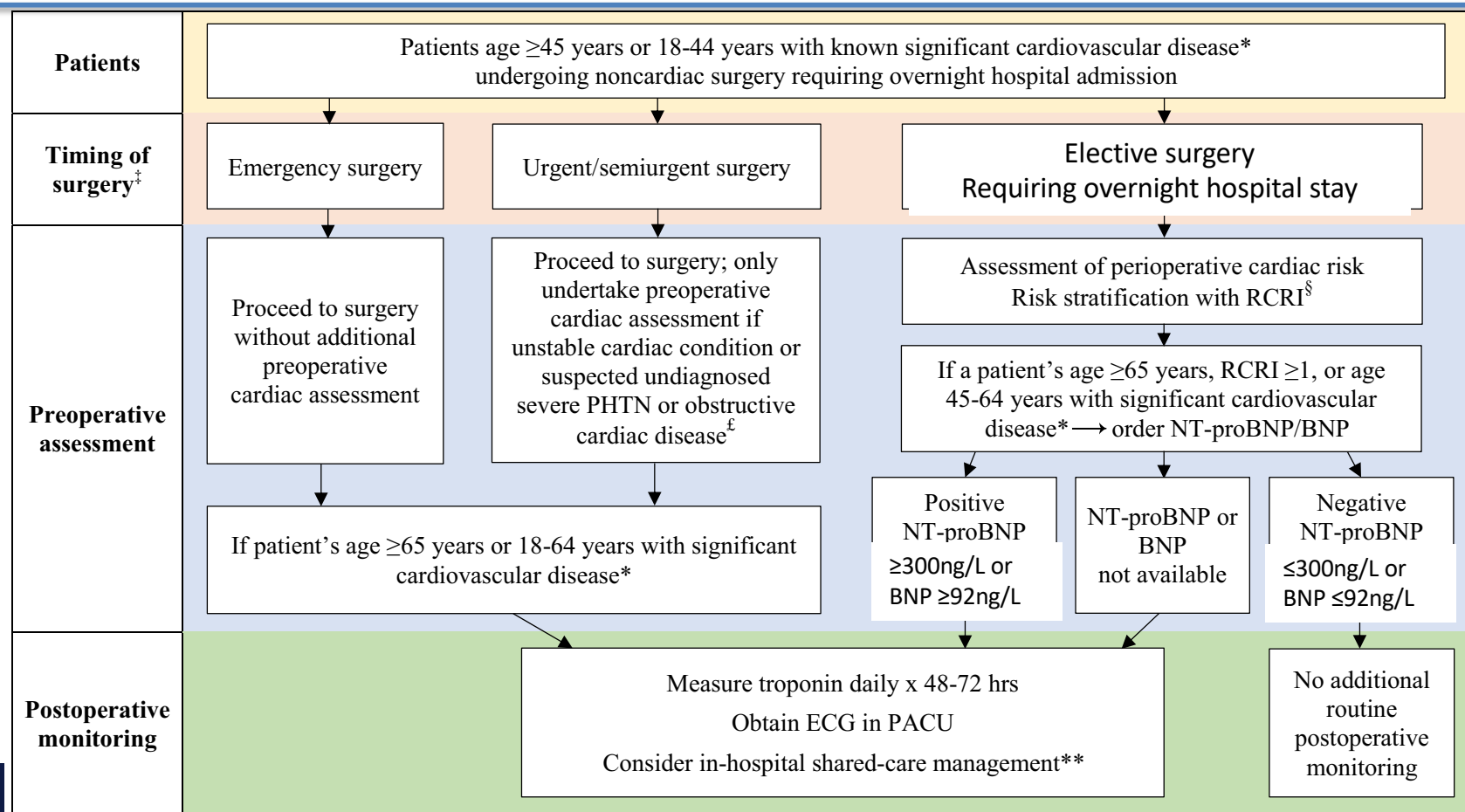
Order a couple select tests

Communicate with patient, surgeon, anesthesiologist

Follow-up perioperatively

- **We DON'T delay emergent/urgent surgery**
 - Exceptions? ... Active ACS, decompensated HF, unstable arrhythmia (things you must emergently optimize)
- **We MAY delay purely elective surgery for medical optimization**
 - Example: patient with poorly controlled diabetes before elective TKA

CCS 2016: Preop Cardiac Risk Evaluation



Emergency surgery

Life or Limb requiring surgery immediately / within hours
(eg) Trauma, AAA rupture, necrotizing fasciitis

Go to OR

(After discussion of goals of care)

If pt ≥ 65 or 18-64 w/ **major CVD** :

-postop Troponin X 48-72h

-PACU ECG

-“shared care management”

Major/Significant CVD=

- Known CAD
- Cerebral vascular disease
- Peripheral Artery Disease
- CHF
- Severe pulmonary HTN
- Intracardiac obstruction
 - severe aortic/mitral stenosis, HOCM

Urgent Surgery

Inpatient surgery within 24-48h (eg) bowel obstruction, hip fracture

Semi-urgent Surgery = Cancer, CEA (urgent outpatient cases)

Go to OR

Consider pre-op tests **only** if you suspect:

- Severe obstructive cardiac condition (ex AS)
- Severe pulmonary HTN
- Unstable cardiac condition - active ACS or arrhythmia

If pt ≥ 65 or **18-64** w/ **major CVD** :

- postop Trop X 48-72h
- PACU ECG
- “shared care management”

Elective Surgery

***in patients $\geq 45y$ or *18-44y w major CVD* requiring overnight hospital stay**

In the Pre-Admission / Pre-Op Clinic:

**Calculate RCRI score
to assess risk of MACE**

Measure **BNP** if:

$\geq 65y$ or

$RCRI \geq 1$ or

45-64y w/ major CVD

BNP Normal

= no additional routine postop monitoring

BNP Abnormal or BNP unavailable:

-postop trop x 48-72h

-PACU ECG

“shared care management”

BNP Cutoffs: NT-proBNP ≥ 300 ng/L or BNP ≥ 92 ng/L

Revised Cardiac Risk Index

Score out of 6, 1 point for any of the following variables

High Risk Surgery	Intraperitoneal, intrathoracic, supra-inguinal vascular
Ischemic Heart Disease	History of MI, positive exercise stress test, current angina or nitrate use, ECG w Q waves. Note: If pt has revascularization w PCI or CABG does NOT get a point in absence of above – e.g. if patient had CABG and is asymptomatic, no nitro, with normal ECG, stress test does not get a point.
CHF History	HISTORY of CHF, pulm edema, PND OR exam showing bilat crackles, S3 gallop OR vascular redistribution on CXR
History of stroke or TIA	Score 0-6 ~ risk of infarct, arrest or death 30d postop 0 ~ 4% 1 ~ 6% 2 ~ 10% 3+ ~ 15%
Diabetes on Insulin	
Pre-op Creat >177 umol/L	

Why the BNP?

VISION NT-proBNP Study

- Prospective cohort study, had pre-op NT pro BNP measured
- 10,402 patients undergoing major (with stay overnight) non-cardiac surgery
- BNP highly predictive of post-operative adverse cardiac events

Preoperative NT-proBNP - VISION Study

	<100 ng/L % (95% CI)	100 to <200 % (95% CI)	200 to <1500 % (95% CI)	>1500 % (95% CI)
Composite of CV death and MINS	5 (5-6)	12 (11-14)	21 (19-22)	38 (34-41)
Composite of total death and MI	2 (1-2)	3 (2-4)	8 (7-9)	16 (13-19)
MINS	5 (5-6)	12 (11-14)	20 (19-22)	36 (32-40)
Cardiovascular death	0.2 (0.1-0.3)	0.4 (0.3-0.7)	0.7 (0.4-1.0)	3 (2-4)
All-cause death	0.3 (0.1-0.4)	0.7 (0.3-1.1)	1 (1-2)	4 (2-6)

Ann Intern Med. 2020;172:96-104

CCS 2016: pre-op BNP testing

Recommendation: Measure NT-proBNP in anyone $\geq 65y$ or RCRI ≥ 1 or 45-64y with major CVD

Rationale: enhance prognostication for high-risk patients without expense, delay & difficulty of other forms of testing

- Based on studies showing BNP is independently associated with MACE and improves risk estimation compared compared to RCRI alone

CCS : pre-op cardiac testing

We recommend **against** performing preoperative stress testing *to enhance perioperative cardiac risk estimation*

Including:

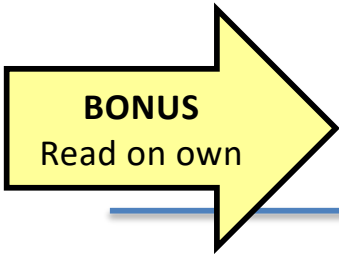
- exercise stress testing
- cardiopulmonary exercise testing (CPET)
- pharmacological stress echocardiography
- pharmacological stress radionuclide imaging



(However, if patient has **SYMPTOMS** you are concerned with (eg) crescendo or unstable angina, consider testing as you usually would in outpatient scenario)

CCS rationale against routine preop testing

- Abnormal results lead to delays, invasive revascularization procedures
- Studies of pre-operative revascularization have **failed to show an improvement in post-op outcomes** (at expense of delay of surgery)
 - CARP (Coronary Artery Revascularization before Major Vascular Surgery- McFallis EO et al, NEJM 2004)
 - delay of median OR by 36days
 - no improvement in post-operative MACE in the revascularization group
 - Meta-analysis of nearly 4000 patients undergoing high-risk noncardiac surgery failed to show improvement in post-operative MACE in coronary revascularization vs medical therapy (Wong EY et al, Can J Anesthes. 2007)



High risk features of non-invasive stress testing

- ≥ 2 mm ST depression
- ST elevation
- VT/VF
- sBP not greater than 120 or decrease >10 mmHg
- EF $< 35\%$
- Severe stress induced LV dysfunction - EF $<45\%$ or decrease with stress by over 10%

Scenario: Patient waiting for elective surgery complains of new exertional retrosternal chest pain. You appropriately order a stress test in this scenario (to investigate symptoms, not to enhance periop risk evaluation). The patient has 3mm horizontal ST depression inferiorly during treadmill stress.
= delay elective OR and refer to cardiology

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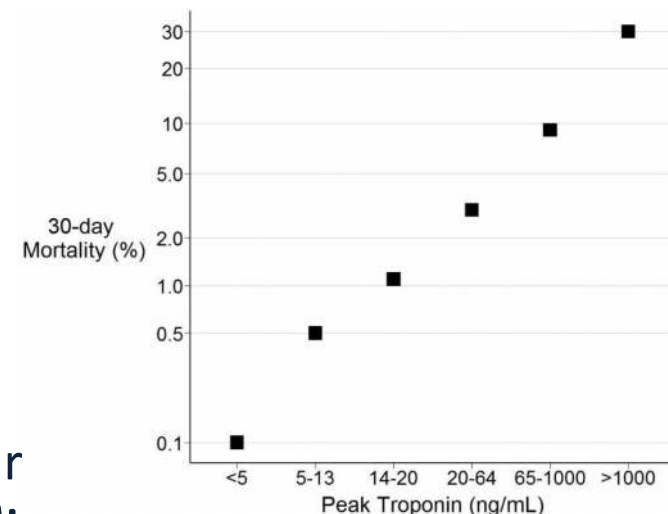
-PACU ECG

“shared care management”

BNP Cutoffs: NT-proBNP ≥ 300 ng/L or BNP ≥ 92 ng/L

Post-Op Troponin – why bother?

- CCS: >65% of patients suffering a perioperative myocardial infarction do not experience ischemic symptoms
- **“MINS” myocardial injury after noncardiac surgery**
 - Acute elevation of Troponin due to myocardial ischemia within 30d of noncardiac surgery
 - » Majority occur in first 72h postop (VISION)
 - Myocardial injury due to ischemic mechanism
 - Other causes excluded – PE, renal failure, sepsis
- Asymptomatic MIs / MINS are associated with a similar increased risk of 30-day mortality (adjusted OR, 4.00; 95% CI, 2.65-6.06) as symptomatic myocardial infarctions
- Incidence of MINS up to 18% (**Cardiol Rev.** 2019; 27:267–273)



VISION study Lancet 2012

BONUS
Read on own

MINS Troponin Cutoffs

- Troponin T (4th generation Tn) > 0.03 ng/mL
- hsTroponin T (5th generation) 20-65ng/L with absolute change >5 ng/L or absolute level >65 ng/L
- A hsTnI ≥60 ng/L (ARCHITECT STAT Abbot assay)⁹ and a hsTnI ≥75 ng/L (Siemens Healthineers ADVIA Centaur Assay)
- For other troponin assays, physicians should consider any elevation above the 99th percentile URL as the threshold for MINS.

MINS Management



- ✓ Clinical assessment, ECG
 - if STEMI consider invasive angiography
 - If high risk features, consider inpatient risk stratification

Criteria for definition of MI met? (Any of: Symptoms of ischemia / new ischemic changes / pathologic Q waves / Imaging evidence of new regional wall motion abnormality / Coronary thrombus on angio/CCTA)

If YES, think if mechanism:

- (i) atherosclerotic plaque disruption (type 1 MI) → consider revascularization, DAPT, high intensity STATIN, BB, ACE/ARB
- (ii) supply-demand mismatch (type 2 MI)

- ✓ Identify nonischemic triggers for troponin rise and treat
 - Anemia, tachycardia, sepsis, PE etc
- ✓ Treat with medications for secondary risk reduction
 - ASA, statin for all
 - Consider dabigatran 110 mg bid?
 - MANAGE trial – postop MINS RCT, but 40% discontinued med in followup
 - Optimize other risk factors: diabetes, smoking cessation, BP



BONUS
Read on own

What DOESN'T work for MINS

Ivabradine (does not PREVENT –MINS (trial, Circulation August 2025))

Colchicine (**Does not ↓MINS or POAF [COP-AF, 2023]**)

TXA (2022 POISE 3 – but it does prevent major bleeding!)

Periop Vit C IV, N-Acetylcysteine, high FiO₂ (80% in OR) don't prevent MINS.

High intraop MAP >80 vs >60 no difference in MINS in adults >45y with CV disease and chronic HTN (POISE3 – Ann Int Med 2023)

Warming intraop to Temp>37°C vs standard 35.5°C in adults >45 w/ 1+ CV risk factor undergoing noncardiac surgery did not reduce endpoint of MINS, arrest, death at 30d. (PROTECT – lancet 2022)

?? Dabigatran 110mg BID in postop MINS reduced rates of MACE but >40% of patients discontinued med in followup and trial terminated early due to slow enrollment. *Not incorporated into guidelines or clinical practice* (MANAGE Trial Lancet, 2018).

Timing of surgery with Recent Angioplasty

(CCS 2018/new CCS 2023 Antiplatelet guideline does not update these recommendations for non-cardiac surgery)

Elective surgery

- Delay 14 days after POBA
[AHA/ACC 2024, no guidance in CCS 2018]
- Delay at least 1 month after BMS
- Delay at least 3 months after DES - If semi-urgent (ex. Malignancy) delay at least 1 month post-PCI - CCS
- Continue ASA wherever possible
[CCS 2018, ACC/AHA 2024]
- Hold clopidogrel / ticagrelor 5-7 days, prasugrel 7-10 days.

* **ACC/AHA 2024 differ slightly (below). But we stick to CCS!**

- * {
 - 12 months if DES for ACS
 - 6 months if DES for Chronic CAD
 - 3 months if risk of delaying surgery > risk MACE

Urgent/Emergent surgery

- Do not delay. Book monitored bed, counsel patient
 - Decision to stop P2Y₁₂i = Talk to patient, cardiologist, anesthesiologist, surgeon
 - Usually surgery by being URGENT means operation in <48h so even if you stop P2Y₁₂i, its effect will be active for 3-5 day = no neuraxial anesthesia
- Restart DAPT after surgery “as soon as deemed safe by surgeon”

Timing of surgery post-Stroke

Association of Time Elapsed Since Ischemic Stroke With Risk of Recurrent Stroke in Older Patients Undergoing Elective Non-neurologic, Noncardiac Surgery

Laurent G. Glance, MD^{1,2,3,4}; Curtis G. Benesch, MD⁵; Robert G. Holloway, MD⁵; [et al](#)

[» Author Affiliations](#) | [Article Information](#)

JAMA Surg. 2022;157(8):e222236. doi:10.1001/jamasurg.2022.2236

Cohort study analyzing time from hospitalization for acute stroke to ELECTIVE surgery. Findings:

- Patients who have had a previous stroke have a higher risk of 30d perioperative stroke/death/discharge to LTC
 - Surgery within 30days: OR 8.02
 - Surgery 30-90d : OR 5.01
 - Surgery 90-360d: OR 4.76

So what's an Internist to do?

ACC/AHA 2024 – “delay elective Non Cardiac Surgery for **≥3 months** after the most recent event to ↓ risk of recurrent stroke”
Re Antiplatelets: “continue ASA for low/moderate risk bleeding procedures / interrupt ASA for high risk surgery/procedure”

Exceptions to timing post-PCI or post-stroke:

Cardiovascular surgery

- Perioperatively, **all patients undergoing elective vascular surgery need to be on an antiplatelet**
 - You won't get asked this – Surgeon Decides - usually ASA 81mg-325mg
 - 1 study (CAPRIE, ~6500 pts) showed monotherapy clopidogrel reduced MACE more than ASA.
 - For non **carotid** surgery, DAPT is not routine. They suggest "ASA, Clopidogrel or ASA + Clopidogrel" for lower extremity bypass – surgeon specific, ask your surgeon.
- **Carotid endarterectomy** [strokebestpractices.ca]
 - Symptomatic extracranial Carotid stenosis –go for urgent revascularization
 - "ASA and P2Yi continued perioperatively"
- **Cardiac Surgery* (GIM exam focus is on NON CARDIAC SURGERY)**
 - CV surgery patients continue ASA periop.
 - P2Yi is usually held if going "on pump" (bypass)
 - Hold Clopidogrel 2-7 days based upon hemodynamic stability, anatomy etc [CCS 2023]
 - Hold ticagrelor 2-3 days prior to CABG rather than 5-7 [CCS 2023]
 - Recommend DAPT be restarted postop "as soon as it's safe"

Society for Vascular Surgery Practice guidelines, 2015.
Strokebestpracticesguidelines.ca 2020 Update
CCS 2023 Antiplatelet Guidelines 25

You Hear This Murmur in Pre-Op Clinic

- SEM RUSB late peaking, absent S2, radiates to clavicle and R carotid with weak carotid upstroke, brachioradial delay, murmur softer with squat → stand...



Credit: <https://teachingheartauscultation.com/heart-sounds-mp3-downloads> - open source self-learning program! Can use for teaching your med students!

RULE IN AS	RULE OUT AS
Slow rise of carotid pulse (LR+2.8-130)	Absence of radiation to R carotid (LR- 0.05 – 0.10)
Mid to late peaking murmur (LR+ 8-101)	
Soft S2 (LR+ 3.1-50)	(Ref: Etchells E et al. JAMA 1997)

Valvular Heart Disease (VHD)

ECHO if: as indicated in any other scenario

- Symptoms of possible severe VHD
- Exam suggests mod-severe stenosis or regurgitation
- Not done in >12m w known advanced VHD

DELAY elective non-cardiac surgery if meets criteria for valve intervention

“Reasonable to proceed” with elective non-cardiac surgery if - *asymptomatic* severe valve disease and low risk surgery – details beyond scope of GIM – consult their cardiologist

Urgent/Emergent OR and suspected severe VHD

- Try to get ECHO if possible – Inform anesthesia, consider postop monitored bed, art line
- “*Team Based Patient Centred Decision*” – ACC/AHA 2024

Low Risk = combined patient and surgical characteristics predict a risk of MACE <1%

Criteria for Valve Intervention to know:

- See Cardio Slides for high yield summaries for AS, MS, MR, AR surgical indications

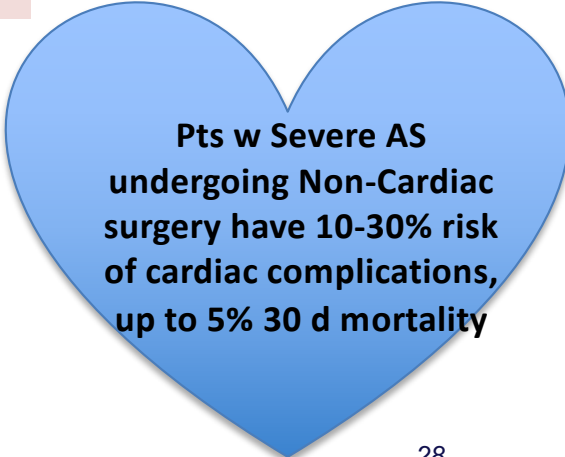
Severe AS on echo

Abnormal systolic AV opening with
 $V_{\max} \geq 4\text{m/s}$ **or**
 $P_{\text{mean}} \text{ gradient} \geq 40\text{mmHg}$

- **Class 1 indications for AVR**

– Class 1 indications to replace:

- Severe AS with symptoms
- OR Severe AS without symptoms, but EF <50%
- OR Severe AS going for other cardiac surgery ex CABG



**Pts w Severe AS
undergoing Non-Cardiac
surgery have 10-30% risk
of cardiac complications,
up to 5% 30 d mortality**

Severe AS on echo
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Severe AS on echo
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- Suspect if: Symptoms of Severe AS
but Vmax only 3-3.9m/s, or (LOW FLOW)
Pmean 20-39mmHg (LOW GRADIENT)

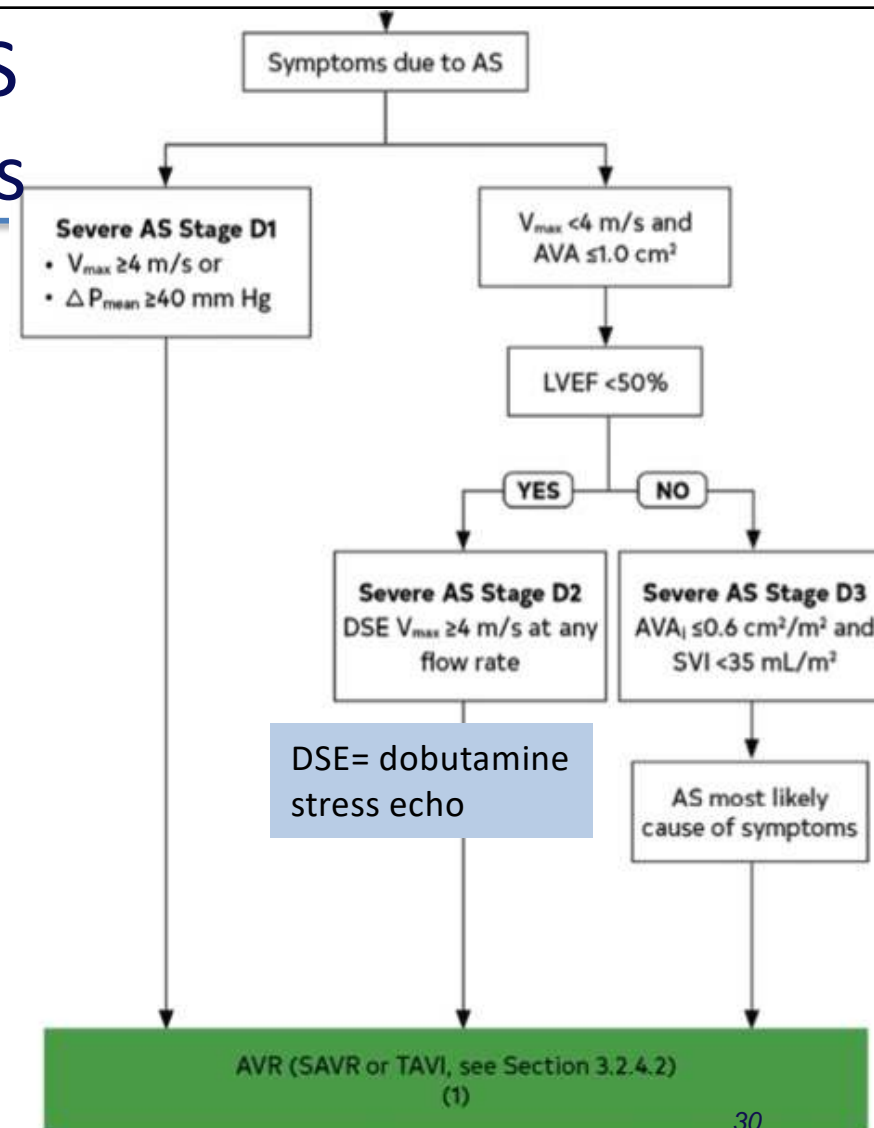
Classically, in patients with reduced LVEF, but can occur with preserved EF and low stroke volume. The low output results in 'pseudonormalization' of gradients and will underestimate AS severity

What to do with “Moderate” AS by ECHO w Severe AS Symptoms

KEY CONCEPTS: low flow / low gradient AS is advanced stage AS + is symptomatic, but does not meet ECHO criteria for severity

- If SYMPTOMATIC + EF <50 = order dobutamine stress echo
 - “Classical low-flow, low gradient” – 5-10% AS!
- If SYMPTOMATIC + EF >50 = AVR if expert believes AS cause of symptoms

2020 AHA/ACC guideline on
Management of Valvular Heart Disease



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Cr 87 (CrCl 41)

NT –pro BNP = 780 ng/L

ECHO – mild aortic stenosis – heavily calcified, mean gradient 18mmHg, peak velocity 2.0 m/s, AVA 1.5 cm². LVEF 50%.



Post-operative Atrial Fibrillation (POAF)

- Refers to de novo Afib diagnosed intraop/postop
 - Associated with increased risk of stroke, mortality and MI 1 year post OP [POISE1 data]
- Reviews but no RCTs of **non-cardiac surgery patients**
 - Suggest POAF $\approx$ nonvalvular AFIB thromboembolism risk
 - Only $\approx$ 20-30% of patients with POAF receive anticoagulants
 - If patient is cardioverted, must anticoagulated per usual at least 4 wks post
 - ASPIRE-AF trial currently recruiting – does DOAC given to treat POAF $\downarrow$ risk of stroke and systemic embolization, vascular mortality?
- Bottom line: Weigh risk benefit on each case
 - Prolonged outpatient monitoring/clinic follow-up prudent
 - No wrong answer on oral exam [“weigh risk/benefits and follow-up”]

ACC/AHA 2024: Treat underlying triggers of AF (sepsis, anemia, pain....)
Follow-up recommended / risk stratification given $\uparrow$ risk of AF recurrence
Initiate anticoagulation after weighing risks / benefits *[text favours anticoagulation]*

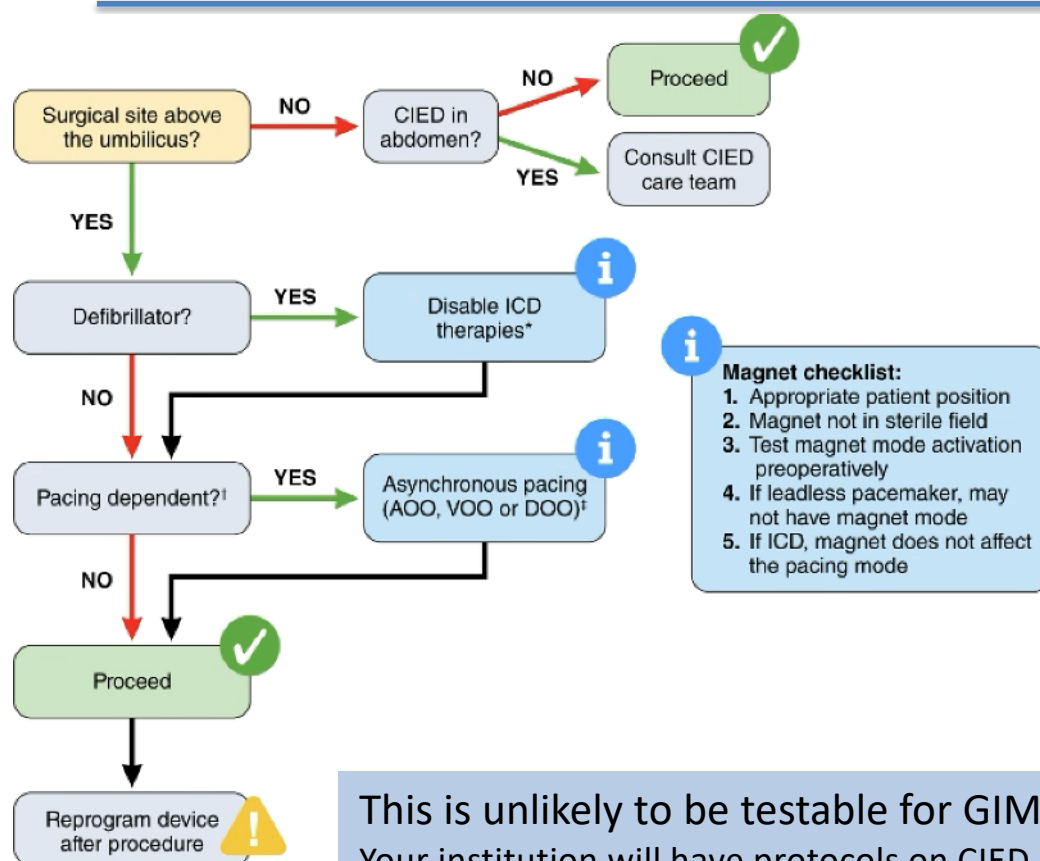
Source: Phri.ca
J Am Coll Cardiol. 2018, 72(17):2027-2036.



Perioperative PPM/ICD Management

AHA 2024 Statement Circulation. 2024;150(8):e183

ACC/AHA 2024 Perioperative Cardiovascular Management for Noncardiac Surgery



• Key points:

- Need to know indication for device, type and location of device
- Type of surgery and cautery used
 - Bipolar cautery does not cause electromagnetic interference but monopolar does
 - Return electrode on contralateral lower limb
- ICD – magnet application inhibits arrhythmia detection but pacing function preserved
- Be aware of leadless devices

This is unlikely to be testable for GIM exam

Your institution will have protocols on CIED management intraoperatively

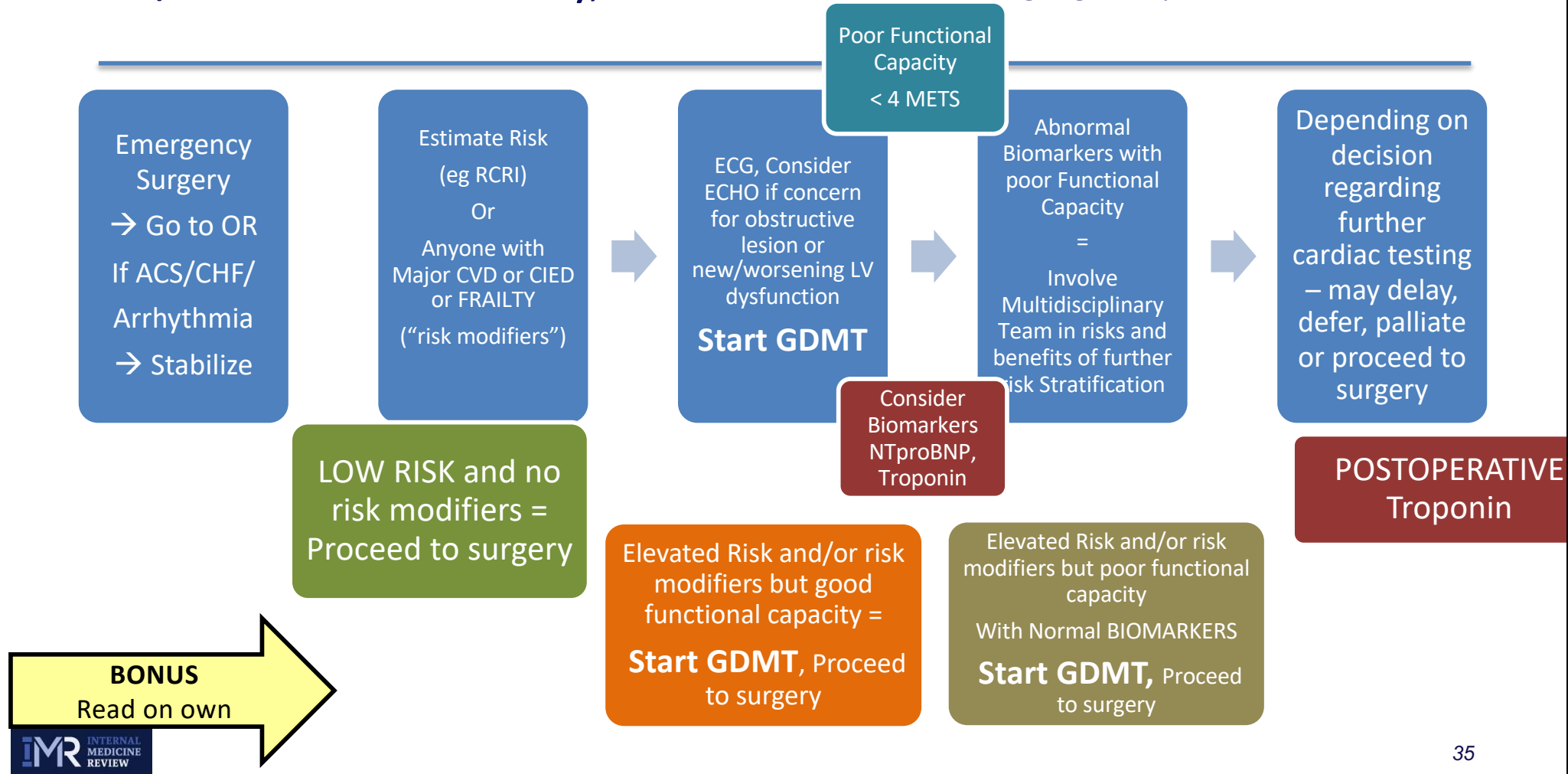
If reprogram of device post procedure is required ensure your facility has capability for this

Peri-op Scenario: Cardiovascular Bottom Line

- **Good history and physical**
 - Don't miss obstructive cardiac lesion!
- **Counsel patient on their patient & surgery specific risk**
 - Use RCRI and history with NT-pro BNP to identify patients who need postop MINS screening
 - Engage in “shared decision making” with patient and surgeon prior to truly elective OR
- **Don't order unnecessary pre-op tests** in asymptomatic/stable patients
 - ECHO if you suspect obstructive lesion or severe valvular heart disease
- Put on OMT where indicated by known cardiovascular conditions
- Plan to monitor patient closely with clinical exam (+/- ECG, Tn postop)

AHA/ACC 2024 – FYI only, stick to CCS

Here to highlight importance of FRAILITY



AHA/ACC 2024 - FYI

CCS

Assess Pre-Operative Risk

✓

*

✓

- Preop assessment of MACE
 - using RCRI or other validated tool
- Preop assessment of Frailty in those >65 or those <64y w perceived frailty
 - Using Clinical Frailty Scale or other validated tool
- Patients ≥65 or ≥45 with symptoms of CVD it is reasonable to measure BNP or NT-proBNP or cardiac Troponin to supplement evaluation of risk

Pre-op Diagnostic tests

X

X

✓

✓

- ECG unless asymptomatic undergoing low risk procedures
- “Consider” Stress testing or CCTA in patients having high risk surgery if patients have poor (≤ 4 METs) or unknown functional capacity for detection of inducible ischemia or high risk coronary anatomy
- ECHO if new symptoms of CHF or worsening of CHF with known LV dysfxn
- BUT – *routine invasive coronary angiogram is not recommended to improve perioperative outcomes*
 - **Caveat – hemodynamically significant Left Main disease*

The Pre-op Assessment

The Checklist:

- ✓ Cardiac risk assessment
 - ☐ + Don't forget endocarditis prophylaxis
- ☐ **Delirium Risk and Frailty**
- ☐ Manage Medications
 - ☐ BB, RAASi, statins, Antiplatelets
 - ☐ Anticoagulants – stopping , bridging
- ☐ Heme considerations:
 - ☐ VTE prophylaxis
 - ☐ Anemia optimization
 - ☐ Coagulopathies (liver, ITP, VWD etc.)
- ☐ Metabolic: Diabetes, stress steroids
- ☐ Respiratory Risk Stratification



Frailty

“Biologic Syndrome of decreased reserve and resistance to stressors, resulting from cumulative declines across multiple physiologic systems, and causing vulnerability to adverse outcomes” (Fried LP et al; 2001)

- AGE does not define Frailty ... FUNCTION does
 - Physical performance, gait speed, mobility, nutritional status, mental health, cognition
 - Useful in Geriatrics to communicate “overall status” of a patient
 - **Identifies patients at greatest risk of declining or dying due to decreased reserve**
- Surgery is a state of physiologic stress
 - Patients with pre-operative frailty have worse outcomes
 - ↑Death, ↑ Delirium, ↓Discharge Home (↑ to LTC and “non-home destinations”)

Surgery is a major stress test!



Am J Surg. 2019 Aug;218(2):393-400.
McIsaac DA Anesth Analg 2020;130:1450-60
Handforth C et al Ann Oncol 2015 Jun;26(6):1091-1101

CLINICAL FRAILITY SCALE

	1	VERY FIT	People who are robust, active, energetic and motivated. They tend to exercise regularly and are among the fittest for their age.
	2	FIT	People who have no active disease symptoms but are less fit than category 1. Often, they exercise or are very active occasionally, e.g., seasonally.
	3	MANAGING WELL	People whose medical problems are well controlled , even if occasionally symptomatic, but often are not regularly active beyond routine walking.
	4	LIVING WITH VERY MILD FRAILITY	Previously "vulnerable," this category marks early transition from complete independence. While not dependent on others for daily help, often symptoms limit activities . A common complaint is being "slowed up" and/or being tired during the day.
	5	LIVING WITH MILD FRAILITY	People who often have more evident slowing , and need help with high order instrumental activities of daily living (finances, transportation, heavy housework). Typically, mild frailty progressively impairs shopping and walking outside alone, meal preparation, medications and begins to restrict light housework.

	6	LIVING WITH MODERATE FRAILITY	People who need help with all outside activities and with keeping house . Inside, they often have problems with stairs and need help with bathing and might need minimal assistance (cuing, standby) with dressing.
	7	LIVING WITH SEVERE FRAILITY	Completely dependent for personal care , from whatever cause (physical or cognitive). Even so, they seem stable and not at high risk of dying (within ~6 months).
	8	LIVING WITH VERY SEVERE FRAILITY	Completely dependent for personal care and approaching end of life. Typically, they could not recover even from a minor illness.
	9	TERMINALLY ILL	Approaching the end of life. This category applies to people with a life expectancy <6 months , who are not otherwise living with severe frailty . (Many terminally ill people can still exercise until very close to death.)

SCORING FRAILITY IN PEOPLE WITH DEMENTIA

The degree of frailty generally corresponds to the degree of dementia. Common **symptoms in mild dementia** include forgetting the details of a recent event, though still remembering the event itself, repeating the same question/story and social withdrawal.

In **moderate dementia**, recent memory is very impaired, even though they seemingly can remember their past life events well. They can do personal care with prompting.

In **severe dementia**, they cannot do personal care without help.

In **very severe dementia** they are often bedfast. Many are virtually mute.



Clinical Frailty Scale ©2005–2020 Rockwood, Version 2.0 (EN). All rights reserved. For permission: www.geriatricmedicine.ca
Rockwood K et al. A global clinical measure of fitness and frailty in elderly people. CMAJ 2005;173:489–495.

Clinical Frailty Scale (Rockwood K, 2005)

- Designed to augment a Complete Geriatric Health Assessment
- Due to ease of use now being validated as a screening tool
 - Ask about BASELINE – if they just broke their hip today, CFS is automatically a 6-7, ask about last week.
 - Chronic conditions don't automatically score a 3 (can still be CFS1-2)
 - Change in function matters (a SAHM who NEVER did the finances wouldn't be scored frail 5 because accountant does taxes after spouse dies)
- Not validated yet for those <65
- Others: Fried Frailty Index

Pre-op Case

82yF living in apartment alone, scheduled for **R hip arthroplasty in 10 days**. She has a history of atrial fibrillation, **HFnrEF** (LVEF 50%), CKD (eGFR 45).

She is now confined to apartment due to severe hip pain. She enjoys card games and plays online bridge with her grandchildren. She manages her meds and finances but has help for meals (meals on wheels), shopping, housekeeping, and PSW to support bathing.

Meds: apixaban 2.5mg po BID, dapagliflozin 10mg daily, furosemide 40mg daily, metoprolol 25 mg po BID, candesartan 16mg daily.

Exam: BP 132/78, HR 80-90 irregularly irregular. 98% room air. Wt: 59 kg.

Systolic ejection murmur crescendo-decrescendo, Normal s1, s2 with radiation to clavicle.

Chest clear, no peripheral edema, JVP not elevated.

Struggles to stand and walk 10m in clinic due to hip pain.

Hgb 119, ferritin 40 / Tsat 15%, plt 280

Cr 87 (CrCl 41)

NT –pro BNP = 780 ng/L

ECHO – mild aortic stenosis – heavily calcified, mean gradient 18mmHg, peak velocity 2.0 m/s, AVA 1.5 cm². LVEF 50%.

Clinical Frailty Scale = 6



Frailty in Perioperative Medicine

Is the Frailty Index be a better predictor of perioperative morbidity/mortality than RCRI Index?

- Interventions targeting frail adults may be helpful to reduce morbidity
 - Multicomponent interventions, comprehensive geriatric assessments, pre/post op exercise protocols
 - HELP program if cognition/delirium risk
 - PRE-HAB – pre-op rehab
 - Nutrition optimization
 - Multimorbidity... CGA/Geriatric assessment perioperatively
 - **SAME PRINCIPLES as delirium prevention (see Geriatrics Lecture)**

Delirium and Elective Surgery

- Validated tools estimate pre-op risk of delirium for elective non cardiac surgery
- Validated screening tools detect delirium post-op (CAM)
- There are interventions that reduce risk of post operative delirium in at-risk patients
 - Pre-op Comprehensive Geriatric Assessment ¹
 - Multicomponent interventions:
Effect of the Tailored, Family-Involved Hospital Elder Life Program on Postoperative Delirium and Function in Older Adults RCT²
 - Addressed patients' orientation, nutrition, hydration, sleep, and mobility, involved family to help. **NNP = 6**

¹[Anaesthesia](#). 2014 Mar;69(3):259-69.

²JAMA Intern Med. Pub. online Oct 21, 2019

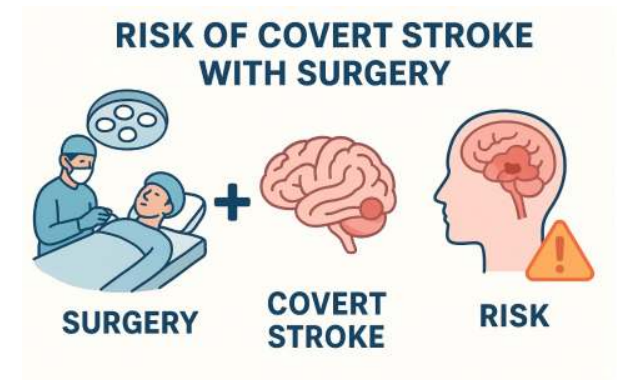
Perioperative delirium and COVERT Strokes

NEUROVISION (2019 Sep 21;394(10203):1022-1029)

- Patients >65y having inpatient, elective non-cardiac surgery
- Postop MRI done during admission
- 7% had covert stroke

PRECISION (Cui Q. et al. Anesthesiology 2025)

- Patients in China >60y having elective non-cardiac surgery
 - *2/3 had neurosurgical craniotomies*
- Postop MRI done within 7 days
- 11.9% had covert stroke
- 3% in non-neurosurgery cases



Development of Covert strokes strongly associated with
Overt stroke or TIA risk at 1 year
Cognitive decline at 1 year

The Pre-op Assessment

The Checklist:

- ✓ Cardiac risk assessment
 - ☐ + Don't forget endocarditis prophylaxis
- ✓ Delirium Risk and Frailty
- ☐ **Manage Medications**
 - ☐ BB, RAASi, statins, Antiplatelets
 - ☐ Anticoagulants – stopping , bridging
- ☐ Heme considerations:
 - ☐ VTE prophylaxis
 - ☐ Anemia optimization
 - ☐ Coagulopathies (liver, ITP, VWD etc.)
- ☐ Metabolic: Diabetes, stress steroids
- ☐ Respiratory Risk Stratification



Medication Management

- Starting and/or stopping:
 - Beta-blockers
 - Antiplatelets
 - Statin
 - RAASi
 - DMARDs & Biologics

MCQ #1 2026

A 44yF presents prior to an urgent thyroidectomy for papillary thyroid cancer. She has a history of hypertension since her pregnancy 4 years ago where she had mild preeclampsia and diagnosed with dyslipidemia postpartum. Exercises regularly without symptoms of concern.

Meds: Candesartan 8mg daily, ASA 81mg (since pregnancy), rosuvastatin 10mg daily.

Exam: BMI 21, 162/89 (BPTru), HR 60 regular, 98% room air

Palpable thyroid nodule Rt lobe. Normal cardiopulmonary exam.

You increase her candesartan to 16mg and add amlodipine 5mg daily. Follow-up BP 140/85.

With regards to her perioperative medication management which is most appropriate in this scenario?

- a) Continue medications as usual on day of surgery
- b) Hold ASA 7 days pre-op, hold candesartan day of surgery, continue other meds as usual
- c) Hold ASA 72h pre-op, hold candesartan day of surgery, continue other meds as usual
- d) Hold ASA 7 days preop, continue other meds as usual

Beta-Blockers

2016 CCS guidelines:

- DON'T start β -blocker within 24 hours of noncardiac surgery (ref: POISE1, 2008)
- DO CONTINUE β -blocker perioperatively if patients taking chronically

THE EXCEPTIONS:

In scenarios where a beta-blocker is clearly indicated (ex. Angina or arrhythmia like rapid AFIB... give beta blockade as you usually would, proceed to OR when appropriate)

Beta-blockade Trial to be aware of



POISE 1 (Perioperative Ischemic Evaluation) Lancet 2008

8351 at-risk patients randomized to receive **100 mg metoprolol XR** or placebo 2-4 hrs pre-op; 30 d outcomes

- **More death (3.1 > 2.3%)**
- **More stroke (1.0 > 0.5%)**
- **Less MI (4.2 < 5.7%)**
 - Most in first week after OR
 - **Most asymptomatic – “MINS” TnI elevation only**

Bottom Line: Don't start high dose beta-blockers on the day of major non-cardiac surgery!

ASA

- We recommend against initiation of ASA for the prevention of perioperative cardiac events
- Physicians should **discontinue ASA at least 3 days** before non-cardiac surgery to reduce the risk of major bleeding.
- **Exception: Continue ASA with history of coronary stent “wherever possible”**
 - unless procedure very high risk bleeding (ex. Neurosurgery)
 - per **CCS 2018/2023 Antiplatelet Guidelines, ACC/AHA 2024 also**

BONUS
Read on own

Why Continue ASA if history of stent?

POISE-2 trial (NEJM 2014)

- 10,010 intermediate-high risk patients randomized 2x2 to ASA or placebo, and clonidine or placebo
 - **4% had stents** – excluded DES <12 mos, BMS <6 wks
 - Excluded anyone on 2nd antiplatelet 72h preop or planned to restart within 7 d postop
- ASA vs placebo
 - Everyone: no antiplatelets for 72h preop.
 - Intervention group: ASA 200 mg given 2-4h before OR, then continued 100 mg OD for 30 days (or switched back to prior dose after 7d)
 - OUTCOME: Equal rate of death/MI, More major bleeding in ASA group (4.6% vs 3.8%)

CONCLUSION: Perioperative use of ASA increases risk of bleeding, no benefit on MI rate.

BUT

POISE-2 Subgroup analysis: **patients w prior PCI had less postop MI if they were int intervention (ASA) arm.**

Several observational cohorts of post-PCI pts support ASA continuation perioperative associated with less 30d MACE. [CCS 2018]

Peri-op Management of other Antiplatelets

Remember the caveat – if indication was a recent PCI, risk of stent thrombosis must be weighed with risk of bleeding (**delay non urgent OR**)

- **Ticagrelor ≈ clopidogrel**
 - hold 5 days preop for most indications
 - hold 7 days for very high bleeding risk
- **Prasugrel**
 - hold 7 days for most indications
 - 10 d for very high bleeding risk - irreversible platelet inhibition

<http://thrombosiscanada.ca/guides/pdfs/Ticagrelor.pdf>

<http://thrombosiscanada.ca/guides/pdfs/Prasugrel.pdf>

Statins

2016 CCS guidelines, 2024 ACC/AHA:

- We recommend continuing statin therapy perioperatively in patients who are on chronic statin therapy
- Patients with MINS should be considered to START statin therapy post-operatively

RAASi (Renin Angiotensin Aldosterone System inhibitors)

2016 CCS guidelines are out of date now. **INDIVIDUALIZE YOUR RECOMMENDATIONS.**

- **OLD Guidance:** We recommend holding ACEI/ARB starting 24 hours before noncardiac surgery in patients treated chronically with an ACEI/ARB. “Restart day 2 after surgery if patient is hemodynamically stable”

Most recent 2024 Studies suggest can either HOLD or CONTINUE, clinical equipoise = individualize:

- POISE3 (Ann Int Med 2024; 176(5):605-614.)
 - Hypotension avoidance (hold RAASi preop) vs Hypertension avoidance (continue all meds periop) = no difference on vascular outcomes or AKI
- Stop or Not (JAMA. 2024;332(12):970-978.)
 - Hold RAASi 72h preop or continue
 - More hypotension requiring pressors in RAASi continuation group but no difference in any MACE/Renal outcomes/postop complications
- **ACC/AHA 2024 $\approx$ individualize**
 - Consider holding if undergoing high risk Non cardiac surgery to avoid hypotension
 - Consider continuing if HFrEF
- What to do? Individualize recommendation on Applied Exam. Unlikely to be an MCQ!
(eg) if planning nephrectomy or high risk AKI = hold

DMARDs and Biologics

“I would consult with the surgeon and rheumatologist”

- Data¹ mixed with regards to ratio risk (↑flare): benefit(↓surgical infection) of holding biologics
- **2022 American College of Rheumatology² guidelines for the Perioperative Management in Pts undergoing elective hip and knee replacement:**
 - ✓ *Nonbiologic DMARDs – continue*
 - x Biologic medications should generally be withheld as close to 1 dosing cycle as able prior to elective THA/TKA & restarted after evidence of wound healing. (eg) infliximab dosed q4 weeks: skip dose, surgery week 5
 - ✓Exception - Severe (organ-threatening) SLE biologics continue w consult of rheum.
 - Restart when wound healing, staples out, ~ 14d

¹J Clin Med. 2024 Sep 18;13(18):5531.

²<https://www.rheumatology.org/Portals/0/Files/Perioperative-Management-Guideline-Summary.pdf>

Practical Tip:

Always remember a patient with RA and neck pain needs a C spine ex-ray prior to the OR to rule out cervical spine subluxation

Pre-OP Med Mgmt – Quick Reference

Drug	Continue periop	Exceptions	Hold Preop (# days)
Beta blocker	✓	Be pragmatic. Hold if ↓ BP,HR	NO
ASA	✗	Coronary Stent Carotid endarterectomy Recent ACS or Stroke	72h or more if very high risk bleed (eg brain/spine surgery)
Clopidogrel or Ticagrelor, Prasugrel	✗	Carotid endarterectomy or lower limb revasc	5d – 7 d 7d – 10d
Statins	✓		NO
ACEi/ARB	individualize	Continue if poor control HTN Hold if high risk AKI, intraop hypotension	24h preop (risk of intraop Hypotension/AKI)
Metformin/DPP4i/SUs	✗		Hold day of surgery
SGLT2i	✗		72h preop
GLP1 Ras	✗	Hold >1 week of surgery	High risk aspiration with induction
Insulin	✓*	Hold prandial when fasting	Dose reduce basal insulin
Nonbiologic DMARDs	✓	Discuss with surgeon, rheumatologist...	Per Periop THA/TKA ACR guideline
Biologics	✗	SLE with organ threatening disease	1 dosing cycle



MCQ #1 2026

A 44yF presents for preop assessment prior to an urgent thyroidectomy for papillary thyroid cancer. She has a history of hypertension since her pregnancy 4 years ago where she had mild preeclampsia and diagnosed with dyslipidemia postpartum. Exercises regularly without symptoms of concern.

Meds: Candesartan 8mg daily, ASA 81mg (since pregnancy), rosuvastatin 10mg daily.

Exam: BMI 21, 162/89 (BPTu), HR 60 regular, 98% room air

Palpable thyroid nodule Rt lobe. Normal cardiopulmonary exam.

You increase her candesartan to 16mg and add amlodipine 5mg daily. Followup BP 140/85.

With regards to her perioperative medication management, what is the best scenario?

- a) Continue medications as usual on day of surgery
- b) Hold ASA 7 days pre-op, hold candesartan 7 days pre-op
- c) Hold ASA 72h pre-op, hold candesartan 7 days pre-op
- d) **Hold ASA 7 days preop, continue candesartan**

D – continue candesartan

POISE 3/ STOP or NOT / AHA 2024

“Individualize recommendations re RAASi perioperatively”

Thyroidectomy: Low risk surgery for bleeding and hypotension

Holding candesartan will result in significant BP elevation

perioperatively (don’t want case cancelled by anesthesia!)

NB - Unclear indication for ASA – can likely “stop forever”, certainly should be held prior to surgery.

The Pre-op Assessment

The Checklist:

- ✓ Cardiac risk assessment
 - ☐ + Don't forget endocarditis prophylaxis
- ✓ Delirium Risk and Frailty
- ✓ Manage Medications
 - ✓ BB, RAASi, statins, Antiplatelets
 - ☐ **Anticoagulants – stopping , bridging**
- ☐ Heme considerations:
 - ☐ VTE prophylaxis
 - ☐ Anemia optimization
 - ☐ Coagulopathies (liver, ITP, VWD etc.)
- ☐ Metabolic: Diabetes, stress steroids
- ☐ Respiratory Risk Stratification



MCQ #2 2026

A 64yF is having a left upper lobectomy for early stage non small cell lung cancer. She has a history of COPD, coronary artery disease with remote MI treated with **LAD stent 12 years ago, and atrial fibrillation**. She had a recent acute exacerbation of COPD treated with antibiotics and steroids for 4 weeks ago. **Surgery is in 3 days**. On exam BP 120/80, HR 55 irreg irreg, SpO2 94% RA, Lungs clear, and labs reveal eGFR >70. No history of CHF, or diabetes.

Medications: apixaban 5mg po BID, metoprolol 25mg po BID, rosuvastatin 20 mg daily, and fluticasone/umeclidinium/vilanterol inhaled daily.

With regards to medications perioperatively, what would you suggest?

- a) Hold apixaban only, continue other medications
- b) Hold apixaban and start asa 81mg daily, continue other medications
- c) Hold metoprolol and apixaban, continue other medications
- d) Hold apixaban, give stress hydrocortisone 50mg IV prior to surgery
- e) Delay surgery as apixaban needs to be held longer

Anticoagulation – General Approach

Procedural Bleeding Risk

Does anticoagulation need to be held? (eg cataract surgery = NO!)

Patient bleeding risk (ie. HAS-BLED score, need for antiplatelet if stent, thrombocytopenia / uremia / cirrhosis)?



Patient Thrombosis Risk

Are they High risk thrombosis if anticoagulants held?

Is bridging needed? **[WARFARIN ONLY]**
If so, how to bridge?

Decision to interrupt antithrombotic therapy for an invasive procedure should balance the risks of a **thromboembolic event** with those of a **periprocedural bleeding event**

- CCS 2020 AFIB Guideline

Procedural Risk: **Low** Bleeding Risk, **anticoagulation can be continued**

- Minor dental procedure:
 - Up to 2 teeth removal; root canal; periodontal surgery; teeth cleaning
 - Continue VKA and use oral prohemostatic agent
- Minor dermatologic procedure: skin biopsy
- Cataract surgery
- Minor procedures with small bore needles (ex paracentesis, thoracentesis)
- Endoscopic procedures not requiring biopsy

Procedural risk: **High** bleeding risk, **must HOLD anticoagulants**

- **Any procedure involving neuraxial anesthesia**
= Regional block, epidural, spinal
- Neurosurgery (intracranial or spinal surgery)
- Cardiac surgery (e.g. CABG, heart valve replacement)
- Major vascular surgery (e.g. aortic aneurysm repair, aortofemoral bypass)
- Major urological surgery (e.g. prostatectomy, bladder tumour resection)
- Major lower limb orthopedic surgery (e.g. hip/knee joint replacement or fracture surgery)
- Lung resection surgery
- Intestinal anastomosis surgery.
- Selected procedures (e.g. kidney biopsy, prostate biopsy, cervical cone biopsy, pericardiocentesis, **colonic polypectomy**)

Who to Bridge? (VKA ONLY**)



- Be CALM and remember Who takes VKA still?
 - Breastfeeding
 - CKD/ESRD with VTE or AFIB
 - APLAS
 - LV thrombus / Likes Warfarin [FRAIL-AF]
 - Mechanical valves

NOT all need
perioperative
bridging!
See next slide!

BRIDGE Trial (NEJM 2015) - Not bridging in patients with **A. Fib on warfarin** was non-inferior to bridging in prevention of arterial thromboembolic events and decreased the risk of major bleeding (NNH = 50).

Limitations: only 3% of patients CHADS 5 or 6

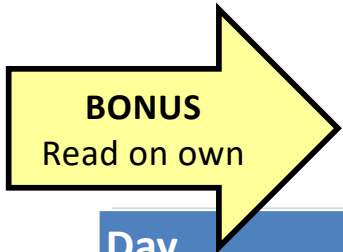
Periop2 Trial (BMJ 2021) – AFIB or mech valve patients given preop bridge; randomized to postop bridge → no benefit to postop bridging (incl >300 Mech Valves, 1/3 mitral)

Bridging



Risk of Thrombosis

Thrombosis RISK	Mechanical Valve (CCS 2016-20, ACCP 2012, AHA/ACC 2020 Valve, ACC/AHA 2024 Periop)	Atrial Fibrillation (CCS 2020)	VTE (ASH 2018)
HIGH RISK (= BRIDGE , weighing in risk of bleeding)	- Mitral mechanical valve - Old AVR (ball-cage, tilting disc) - Mechanical valve with AFIB, prior embolism on anticoag, LV dysfunction, hypercoagulable state. - Stroke w/in 6 mos	CHADS 5, 6 Rheumatic valve Stroke w/in 3 mos Mechanical valve	VTE within past 3 mos High risk thrombophilia • APLA, Prot C/S/ATIII def, or multiple
MODERATE (= No Bridge)	Bileaflet AVR with risk factor • CHADS points but no AFIB, no embolism, no LV dysfxn, no hypercoagulable state	CHADS 3-4	VTE within past 3-12 mos Other thrombophilia Active Cancer (w/in 6 mos) Recurrent VTE
LOW RISK (= No Bridge)	Bileaflet AVR, no afib, no risk factors (Class I ACC/AHA 2020)	CHADS 0-2 (without history of stroke)	VTE > 1 year ago No other risk factors



Example of How to Bridge – Warfarin and LMWH (assumes normal CrCl)



Day		
-5	Stop Warfarin	
-4		
-3	Start full dose LMWH (eg enoxaparin 1mg/kg sc bid)	
-2		
-1	Last dose of LMWH 24h before OR, administer ½ the dose	
Day of OR	}	Start warfarin (1.5X usual dose) at least 12h after OR and full dose LMWH 24h after OR
POD1		
POD2		Warfarin (1.5X usual dose) + LMWH
POD3		Warfarin (usual dose) + LMWH. Check INR and continue LMWH until INR therapeutic then stop LMWH

When to hold OAC : ELECTIVE surgery

Anticoagulant	Bleeding Risk	Day -5	Day -4	Day -3	Day -2	Day -1	<u>SURGERY</u>	POD #1
Anti Xa Drugs* Apixa, Edoxa, Rivaroxa AND Dabigatran**CrCl >50	Low / Moderate	✓	✓	✓	✓	✗	✗	✓ 24h postop
	HIGH Risk	✓	✓	✓	✗	✗	✗	✗
* Assumes CrCl >30 for Anti Xa Drugs. If CrCl is reduced due to AKI – “hold an extra day” ** Assumes CrCl > 50 for Dabigatran. If it is <50 see row below								
Dabigatran With CrCl <30-49	Low / Moderate	✓	✓	✓	✗	✗	✗	✓ 24h postop
	HIGH Risk	✓	✗	✗	✗	✗	✗	✗
Vitamin K Antagonist VKA = warfarin/coumadin	Not Bridging	✗	✗	✗	✗	INR ✗	✓VKA > 12h Postop	✓ VKA
	Bridging	✗	✗	LMWH	LMWH	LMWH ½ dose 24h preop	✗ no LMWH ✓ VKA >12h postop	✓ VKA ✓LMWH if bleeding risk OK

When to restart DOAC post-op

- Moderate risk bleeding surgery
 - Generally resume 24 h postop
- High risk bleeding surgery
 - Resume 48-72h postop
 - Consider alternative anticoagulation until resume DOAC
 - DVT prophylaxis if risk of thrombosis
 - IV heparin or LMWH if high risk thrombosis

Ex. cancer surgery and a <3 month DVT

Moderate Risk: [think: DAY SURGERIES]

General surgery without anastomosis
(eg breast, chole, hernia repair, not cutting bowel)
Arthroscopic Ortho procedures
Multiple dental extractions (>2)
GI endoscopy with biopsies/polypectomy
Non cataract Ophtho surgery



Modified from Schulman and Crowther, Blood 2012;119:3106-3023 and 2020 Thrombosis Canada – Peri-operative Managements of patients who are receiving a New Oral Anticoagulant

DOAC monitoring

[About Us](#) [Clinical Guides](#) **[Tools](#)** [Resources](#) [Education](#)

TOOLS

Algorithms
Anticoagulant Dosing In Atrial Fibrillation
Perioperative Anticoagulant Management Algorithm
Acute Management Algorithms
Atrial Fibrillation
Deep Vein Thrombosis
Pulmonary Embolism
Calculators
CHADS2 Score for Atrial Fibrillation Stroke Risk
CHA2DS2-VASc Score for Atrial Fibrillation Stroke Risk

Perioperative Anticoagulant Management Algorithm

Procedural Bleeding Risk

☐ Low (minor non-dental procedure) ?

☐ Low (minor dental procedure) ?

☐ Moderate ?

☐ High ?

Reset

ThrombosisCanada.ca

Gladstone, DJ, et al. *Ann Intern Med.* 2015;163(5):382-385.

Neuraxial anesthesia and anticoagulation

Anticoagulant	Time BEFORE puncture /catheter manipulation or removal	Time AFTER puncture /catheter manipulation or removal
Warfarin	5 d (check INR, ≤ 1.4)	After catheter removal
IV Heparin	2-4 h (check aPTT)	1h
Heparin sc prophylaxis	4-6h	1h
LMWH - prophylactic	12 h	4h
LMWH - therapeutic	24 h	4h
Dabigatran*	3-5 d(depends on CrCl)	6h
Rivaroxaban/Apixaban/Edoxaban*	3 d	4-6h
Clopidogrel and Ticagrelor	5-7 d	6h
Prasugrel	7-10 d	6h
ASA, NSAIDs	Can continue	Can continue
Fondaparinux (2.5mg)	36-42h	6-12h



VTE Prophylaxis Perioperatively

- Lower yield topic generally (may have 1 MCQ):
 - Take Note of special situations: SPINE AND BRAIN
 - Also take note of dose changes for meds:
 - DOACs all different dosing for PROPHYLAXIS e.g. Rivaroxaban **10** mg po OD
 - If Obese wt >100kg Thrombosis Canada gives “double dose LMWH” (give same dose but BID ex dalteparin 5000 bid, enoxaparin 40 bid)

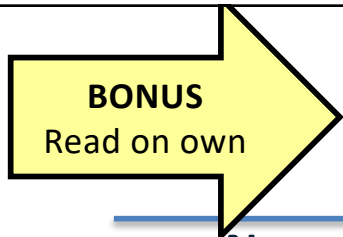


Sources:

- ASH Prophylaxis in Surgical Patients guideline (Dec 2019) (<http://www.hematology.org/VTE/>)*
 - This guideline mirrors Thrombosis Canada
 - Thrombosis Canada suggests every institution have its' own protocol
 - European guidelines (2024) and NICE guidelines (2019) similar

Postop ORTHOPEDIC VTE Prophylaxis

OR Type	Pharmacologic Rx	Duration	Preferences
THA or TKA 	DOACs* > LMWH, fonda *DOAC = Choose any one of Apixaban 2.5 bid, dabigatran 220mg OD , rivaroxaban 10mg OD <i>Edoxaban not licensed in Canada for this indication</i> **Rivaroxaban 10mg x 5 d then ASA 81mg x 9-30d an optional regimen	14-35d Extended prophylaxis (>3 weeks) recommended over short (up to 14 d)	ASH: DOACs are preferred . If DOAC not used, LMWH preferred over VKA. **Do not use ASA alone in patients with additional risk factors (CRISTAL Trial JAMA 2022 –ASA 100 inferior to enoxaparin, does not reduce bleeding either)
Hip Fracture Surgery 	LMWH Fondaparinux LDUH (e.g. if CrCl<30)	14-35 d Extended prophylaxis (>3 weeks) recommended over short (up to 14 d)	LMWH or LDUH recommended over all other agents. DOACs are not studied or recommended for this indication!
Major Trauma/leg amp/admitted	LMWH LDUH (e.g. if CrCl<30)	Until Discharge including Rehabilitation	IPC up front if high risk of bleeding
Notes	Lower leg below knee surgery - Suggest no prophylaxis rather than pharmacologic prophylaxis, unless inpatient – then give ppx until discharged. Arthroscopic surgery – no ppx, unless high risk (eg. Prior VTE, complex repair, cancer). <i>ASH 2019, Thrombosis Canada 2025 White Paper</i>		



Postop VTE Prophylaxis (PPx)

ASH and Thrombosis Canada

- **Non orthopedic** major (inpatient) surgery
 - Generally use pharmacologic ppx. with LMWH. Exceptions below.
 - If high risk bleeding, use IPC (intermittent pneumatic compression) – NOT “compression stockings”.
NO IVC filters for prophylaxis.

Cardiac Sx or Major Vascular Surgery	- <i>Suggests</i> use pharmacological ppx or no pharmacological ppx (paucity of evidence).
Neuro Sx procedures	- Against using pharmacologic ppx (except in high risk such as cancer spine surgery, prolonged immobility post op). - Use mechanical ppx (IPC). *When bleeding risk ↓ post-op, use LMWH or LDUH*
Urologic	- TURP, prostatectomy (assuming robotic assisted/not open prostatectomy, no extended lymph node dissection): Against using pharmacologic ppx
Gynecologic	- Use pharmacological prophylaxis over no pharmacological prophylaxis
Major Trauma	- Low to moderate bleeding risk: Use pharmacological prophylaxis over no pharmacological ppx - High bleeding risk: Against pharmacological prophylaxis initially . Use mechanical (IPC).
Laparoscopic chole.	- Against using pharmacologic ppx (day surgery usually!)
Other General surgery	- Ex Cancer surgery – pharmacologic with LMWH, consider extended ppx for 30 days.

BONUS
Read on own

Postop VTE Prophylaxis Regimens

Non-Pharmacologic

- Graduated compression stockings NOT HELPFUL in addition to LMWH (GAPS study)
- Intermittent pneumatic compression (**IPC favoured over stockings in guidelines**)

Pharmacologic

- LDUH 5000 u sc BID or TID
- LMWH
 - Enoxaparin 30mg sc BID or 40mg sc OD
 - Dalteparin 5000u sc OD
 - Tinzaparin 4500u sc OD or 75u/kg scOD
 - Nadroparin 38u/kg day 1-3 then 57u/kg >POD 4

Pharmacologic (cont'd)

- Fondaparinux 2.5 mg sc OD
- Warfarin (INR 2-3)
- DOACs* for THA/TKA, not for Hip #
 - Rivaroxaban 10 mg po OD
 - Apixaban 2.5 mg po BID
 - Dabigatran 220 mg po OD (150mg po OD if moderate renal impairment)
- ASA (81mg OD POD#6-15 TKA POD#6-35 THA following Xarelto 10mg daily POD #1-5)



***Not all DOACs:**

Edoxaban is not indicated/licensed in Canada for thromboprophylaxis... just for AFIB, treatment of acute DVT/PE

MCQ #2 2026

A 64yF is having a left upper lobectomy for early stage non small cell lung cancer. She has a history of COPD, coronary artery disease with remote MI and coronary stenting 12 years ago, and atrial fibrillation. She had a recent acute exacerbation of COPD treated with antibiotics and steroids for 4 weeks ago. Su HR 55 irreg irreg, SpO2 94% RA, Lungs clear, and CHF, or diabetes.

Medications: apixaban 5mg po BID, metoprolol 2 and fluticasone/umeclidinium/vilanterol inhaled

With regards to medications perioperatively, wh

- a) Hold apixaban only, continue other medication
- b) Hold apixaban and start asa 81mg daily, co**
- c) Hold metoprolol and apixaban, continue ot
- d) Hold apixaban, give stress hydrocortisone 5
- e) Delay surgery as apixaban needs to be held

Answer B.

Apixaban hold Day -2, Day -1, Day of surgery (surgery in 3 days so this is feasible)

Given remote stent, can give ASA 81mg while off apixaban. Continue metoprolol, statin, ICS/LAMA/LABA perioperatively.

Steroids for AECOPD were 4 weeks ago likely short course, always good to think about stress steroids but not required here.

This patient will need postop Troponins, ECGs and planned follow-up for anticipated MINs. Will have dvt ppx postop, restart apixaban ASAP (stop ASA). If you had more time, "PRE-HAB" with RT, Physio, pulm rehab would've been great to suggest here.

Pre-op Case

82yF living in apartment alone, scheduled for **R hip arthroplasty in 10 days**. She has a history of atrial fibrillation, **HFnrEF** (LVEF 50%), diabetes, CKD (eGFR 45).

She is now confined to apartment due to severe hip pain. She enjoys card games and plays online bridge with her grandchildren. She manages her meds and finances but has help for meals (meals on wheels), shopping, housekeeping, and PSW to support bathing.

Meds: apixaban 2.5mg po BID, dapagliflozin 10mg daily, furosemide 40mg daily, metoprolol 25 mg po BID, candesartan 16mg daily.

Exam: BP 132/78, HR 80-90 irregularly irregular. 98% room air. Wt: 59 kg.

Systolic ejection murmur crescendo-decrescendo, Normal s1, s2 with radiation to clavicle.

Chest clear, no peripheral edema, JVP not elevated.

Struggles to stand and walk 10m in clinic due to hip pain.

Hgb 119, ferritin 40 / Tsat 15%, plt 280

Cr 87 (CrCl 41)

NT –pro BNP = 780 ng/L

ECHO – mild aortic stenosis – heavily calcified, mean gradient 18mmHg, peak velocity 2.0 m/s, AVA 1.5 cm². LVEF 50%.

Clinical Frailty Scale = 6





Case resolution

- Goals of care discussion. “Cannot continue to live in pain with limited mobility.” DNR established should anything catastrophic happens. Shared care decision to proceed with case. Safety = request Step down postop, ECGs, tropo ordered, IM consult service to follow-up postop
- “PRE-HAB” referral , arrange for postop care (respite care?) given frailty to support recovery
- *IV iron for anemia*, consider TXA preop/intraop
- DVT ppx postop – can resume usual apixaban POD#1

Drug	Day -4	Day -3	Day – 2	Day – 1	Day of surgery	POD#1
dapagliflozin	take	HOLD	HOLD	HOLD	HOLD	IM to assess
Metoprolol	Take as usual					take
Furosemide	take	Take	take	take	Hold	IM to assess
Candesartan	Take	Take	Take	Hold at least 24h	Hold	IM to assess (likely hold)
Apixaban	Take	Take	HOLD	HOLD	HOLD	Restart! Covers dvt ppx



The Pre-op Assessment

The Checklist:

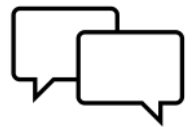
- ✓ Cardiac risk assessment
 - ☐ + Don't forget endocarditis prophylaxis
- ✓ Delirium Risk and Frailty
- ✓ Manage Medications
 - ✓ BB, RAASi, statins, Antiplatelets
 - ✓ Anticoagulants – stopping , bridging
- ☐ **Heme considerations:**
 - ✓ VTE Prophylaxis
 - ☐ **Anemia optimization**
 - ☐ *Coagulopathies (liver, ITP, VWD etc.) – bonus slides, Heme talk*
- ☐ **Metabolic: Diabetes, stress steroids**
- ☐ **Respiratory Risk Stratification**



Anemia and Blood Product Management

- Preop optimization of Hgb if anemic (women: <115, men: <130)
 - Iron deficiency without anemia confers higher risk of postop transfusion (Circulation. 2024;150(8):e183)
 - Consider **pre-operative iron (oral or IV) and erythropoietin**
 - Erythropoietin Stim Agents (EPO) with or without iron reduces the # of pts and # of units of red cells transfused
 - Strongly recommend administration of iron to iron-deficient pts if time permits
 - Target Hgb before major joint arthroplasty (M+F) = 130
 - Review and d/c anti-coagulants as able (see earlier slide deck)
- Tranexamic acid reduces perioperative blood loss, # of patients requiring transfusion and # of units transfused.
 - TXA does not increase risk of thrombosis in perioperative, trauma patients
 - “Reasonable to give TXA to ↓blood loss / transfusions intraoperatively” – ACC/AHA 2024
- More in the wheelhouse of anesthesia but be aware of use of :
 - Acute normovolemic hemodilution to reduce # of pts transfused for cardiac, ortho, thoracic, liver sx
 - Intraoperative Red Blood Cell Recovery (“Cell Saver” device) may reduce # of allo-units transfused
- Consult Hematology for orders for pts with known bleeding disorders!

All-Star Talking Points:
POISE3 Trial – RCT of TXA
vs placebo (NEJM 2022)



Practice Guidelines for Perioperative Blood Management. Anesthesiology 2015; 122(2): 241-00.

Diabetes Management Perioperatively

Uncontrolled diabetes impacts negatively on surgical outcomes

- Increased surgical site infections
- Increased hospital readmissions [major joints]

Perioperative hyperglycemia associated with

- Increased re-operations and infections
- Increased Death

For ELECTIVE Surgery

- “Reasonable” ✓A1C if not done within 3 months if pt has or is at risk of diabetes (ACC/AHA 2024)
- Consider delaying elective major surgery to optimize diabetes control, improve perioperative outcomes
- “In [pts] with poorly controlled DM, **we conditionally recommend delaying TJA to improve glycemic control.**”
 - Set A1C or specific targets not given [ACR Guideline on Optimal Timing of total Joint Arthroplasty (2023)]

Canadian Suggested Standards by CDA working group:

- Institutions should have protocols for pre-op, intra-op and postoperative glycemic management
- **Perioperative BG target 5-10** for most procedures

Can J Diabetes 46 (2022) 99-10.

Diabetes Management Intra-Op: Who gets a Drip?



BONUS

Read on own

- Tight glycemic control perioperatively does not reduce mortality or infections compared to standard control [Bellon F et al. Cochrane Database Systematic Reviews, 2023]
 - ↑ risks of hypoglycemic events
 - Subgroup analysis: *may* reduce CV events (low certainty evidence)
- Who needs a drip? (minimal guidance from Diabetes Canada 2018)
 - Major surgery/CABG should have IV insulin infusion: target BG 5.5-11.1
 - Minor/moderate surgeries: perioperative BG target of 5-10
 - Hypoglycemia protocols should be in place
 - Other instances where IV insulin should be used:
 - Type 1 DM (Any procedure >1-2 hours)
 - If on PUMP consider keeping on pump if have clear institutional protocol to support this, orders from endocrinologist
 - Prolonged OR in insulin dependent patient
 - Intrapartum – diabetics on insulin in pregnancy (see OB Med Lecture)

Diabetes Management Peri-Op: Medication Management



Home Diabetes Regimen	Perioperative Regimen
Oral agents	Hold morning of OR, use sliding scale
GLP1 agonists	Hold/skip weekly injection 1 week prior to surgery **
SGLT2 inhibitor* risk of periop DKA!	Hold 3 days before OR [FDA – 4 days for ertugliflozin]
Long-acting insulin Lantus/Tuojeo, Tresiba, Levemir,NPH	1/2 to full usual dose [70-100% dose if night before surgery injection, but ½ dose if morning of surgery injection]
Twice daily (mix insulin, eg 30/70)	Generally hold, if T1DM need another basal am of OR
3, 4, 5 injections daily	Omit short acting, reduce basal dose as above
DISCLAIMER – this is a guideline. Ask 10 internists and you'll get 10 different answers. Consider risks of hypoglycemia: age, renal fxn, “brittle”/”labile” sugars, etc. You will not get asked to counsel a patient in high detail on periop glucose mgmt due to lack of evidence base!	

Modified from Dhatariya K et al: Management of adults with diabetes undergoing surgery and elective procedures: improving standards. NHS Diabetes April 2011.

*(1) Can J Anesth. 2018; 65: 188-93; (2) AACE/ACE Position Statement. 2016; 22: 753-762

**Am soc Anesthesia 2023: Increased aspiration risk due to delayed gastric emptying; hold 1 week preop. If urgent OR, intubate w aspiration precautions

Resp Risk Assessment

- Smoking Cessation strongly recommended (CCS 2016)
 - >4 weeks cessation shows increased benefit, but anytime you can get someone to quit is better than never!
 - **Joint Replacement:**
 - **“We Conditionally recommend delaying arthroplasty for nicotine use reduction/cessation”** (2023 ACR Guideline on Optimal Timing of total Joint Arthroplasty (2023))
 - Smokers have 1.5-3.2X ↑ risk of infection; ↑ length of stay, ↑ re-operations and worse pain control post-op [American Association of Hip and Knee Surgeons]
- No role for routine pre-op CXR, PFTs *for non-thoracic surgery*. Except if...
 - If result would affect mgmt – e.g. suspected asthma, never tested
 - If Anesthesia requests for neuromuscular disease (predicts need for postop ventilation)
 - If one lung ventilation planned.
 - Lung resection surgery.
 - **Risk score based upon POSTOP PREDICTED FEV1 and DLCO (if both >60%, standard risk)**
 - **No “exact cut-off of FEV1” for lobectomy/pneumonectomy...** It is more nuanced [not testable in MCQ, and NOT for GIM, it is for thoracic surgeon to know!]

BONUS
Read on own

Management of Resp Risk

- Interventions:
 - Pre-op:
 - Optimization of COPD/Asthma (continue puffers)
 - **Smoking Cessation:**
 - **Strongly recommended** (CCS 2016)
 - > 4 weeks cessation shows increased benefit
 - Recent URTI – no good evidence to inform mgmt (delay if active infection)
 - Route of anesthesia – to discuss with Anesthesia non-GA options
 - Post-op:
 - Note: Weak evidence and modest effect for all of these...But easy to mention!
 - Deep-breathing exercises or incentive spirometry
 - Early mobilization
 - Good pain control (esp. upper abdominal surgery – epidural anesthesia is your friend!), NG Decompression
 - DVT prophylaxis
 - Bring Home CPAP machine to hospital and use it on ward!

BONUS
Read on own

Resp Risk Assessment

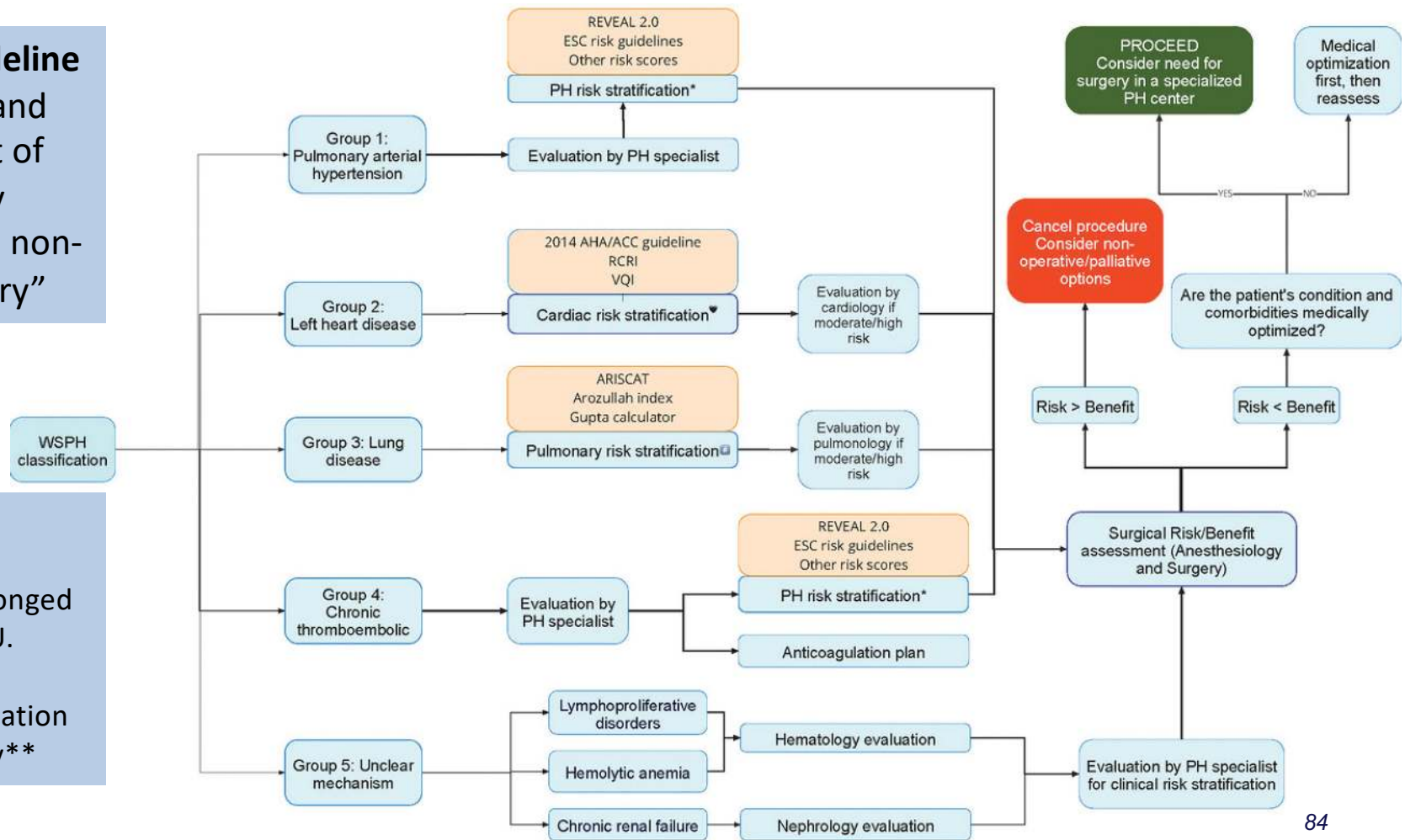
- Less data vs. cardiac risk assessment
 - **Most guidelines have to do with risk of resp failure (eg intubation prolonged)**
- Complications:
 - Respiratory failure, Infection, Atelectasis, Exacerbation, VTE, pneumothorax
- Patient related risks:
 - Clinical=COUGHS (COPD, Older than 60, Underlying disease- ASA >II, General health (functional class), Heart failure, Smoking)***ACP 2006**
 - Biochemical: Albumin <35 (nutritional status)***ACP 2006**
 - Also: Obesity, Asthma, OSA*, Pulmonary HTN*
- Procedure related risks:
 - Surgical site, duration of surgery, route of anesthesia, type of neuromuscular blockade

Pulmonary Hypertension = extra high risk

AHA 2023 Guideline
 “Evaluation and management of pulmonary hypertension in non-cardiac surgery”

Increased risk of mortality, MACE, arrhythmia, prolonged vent support, ICU.

****Preop identification by Internist is key****



Pulmonary Hypertension = extra high risk

- Assessment of risk:benefit requires shared care discussion of surgeon, anesthesia, resp/PH specialist.
- Pulmonary Hypertension in GIM Clinic
 - = “Pump the brakes” on ELECTIVE OR and get more info
 - Internists should be able to identify, workup, refer GROUP 2 and GROUP 3 patients (Pulm HTN due to L heart disease or lung disease).
 - All the complicated (right heart failure) PH patients get seen by a specialist before proceeding with elective surgery
 - Surgery should be at PH centre with experienced anesthesia, access to intraop TEE and postop ICU



Royal College = SAFETY EXAM – if you detect pulm HTN this is clue to workup cause of this and refer prior to non-urgent OR.

Good luck on your exam!!

- BONUS Slides
 - Perioperative case – MINS practice!
 - Perioperative steroid management
 - More on Perioperative blood product management
 - OSA perioperative management
 - Surgery in special populations
 - Pregnancy, Cirrhosis, ESRD
(ACG Perioperative Cirrhosis guideline 2025)
 - COVID perioperatively
 - BONUS MCQs (more online on Flexiquiz!)





Perioperative Medicine Training Program



For inquiries, please
Contact program director
Dr. Vikas Tandon
tandonv@hhsc.ca

FAQs

1) Who is eligible to apply?

PGY4 or above in internal medicine or subspecialties, anesthesiology.

2) How long is the training program?

Between 3 to 12 months.

3) How much time is devoted to clinical vs research interests?

Varies depending on interests, experiences and career goals and can be discussed with the program director in detail.

A 2-year MSc in Health Research Methodology is available for trainees who wish to pursue the degree concurrently.

4) Is funding available?

Trainees are generally funded by their home university. Letters of support are provided for grant applications upon request for the successful applicants.

Mr. Uh-Oh

71y for radical cystectomy and L nephrectomy, non-cancer reasons (chronic infection) at community hospital.

- PMHx T2DM, Obesity (BMI 38), HTN, Remote Ex smoker
- Lives with his wife. Independent for ADLs, most IADLs (doesn't do snow removal – too short of breath). On specific review he is quite sedentary.
- Meds: Semaglutide, Empagliflozin, Metformin, Perindopril/Indapamide, Rosuvastatin, Tamsulosin, Vitamins. Multiple courses of recent antibiotics.
- On exam: 148/92, HR 68, 98% on RA, Lungs clear, Normal heart sounds, no edema, obese abdomen.

Labs

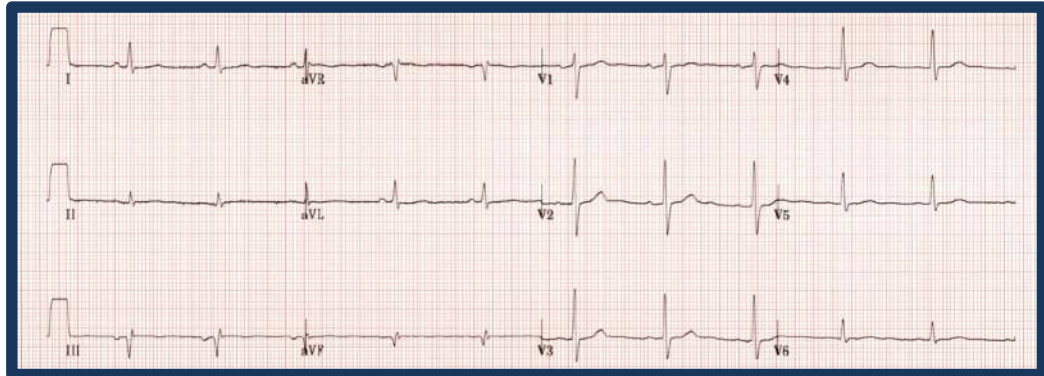
Hgb 114, WBC 8, Plt 250

Tsat 15%, ferritin 29

Na 141, K 4.8, Cr 142

NT pro BNP 458

A1C 7.8%



How would you optimize Mr. Uh Oh to *reduce perioperative CV risk* following major non-cardiac surgery?

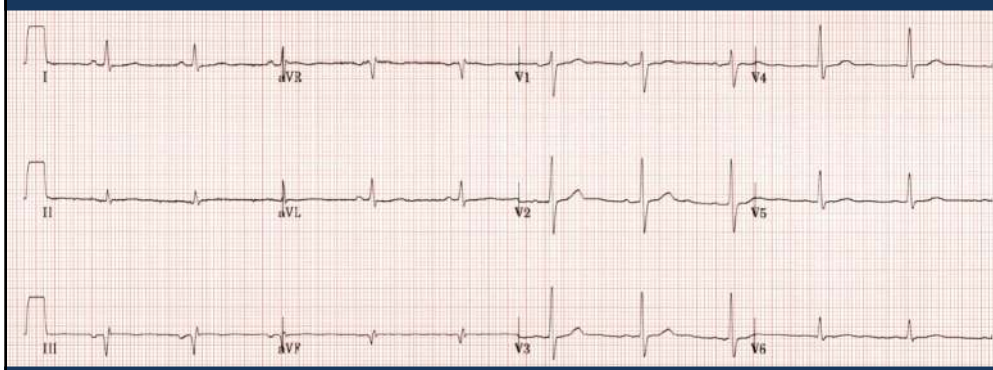
Cardiac: High risk. Postop MINS screening, counsel pt. Consider adding another medication for BP as you will have to hold acei/TZD day of surgery. No role for stress testing or ECHO.

Heme: IV iron! Prior to surgery.

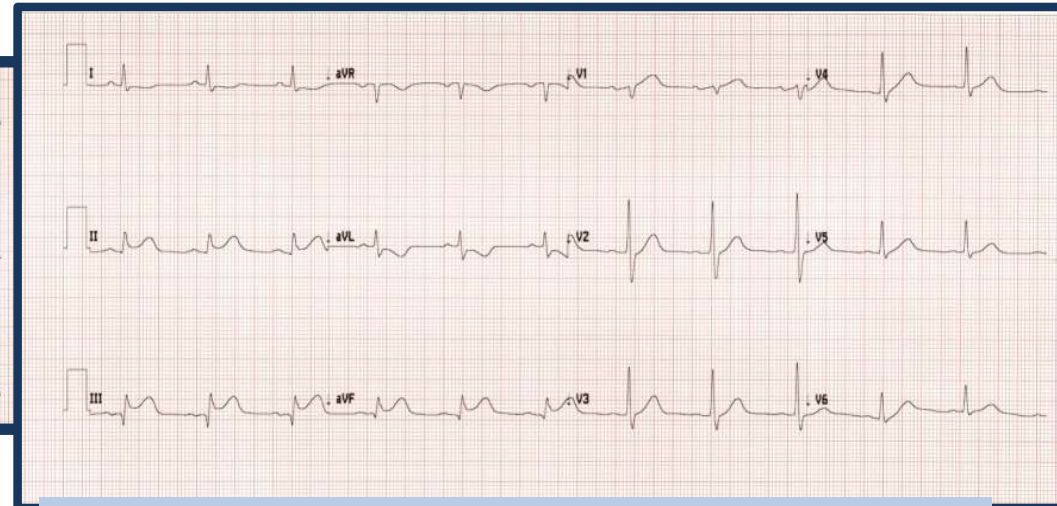
Med reconciliation:

Drug	Day -4		Day -3	Day – 2	Day – 1	Day of surgery
Semaglutide	Skip week of surgery (per anesthesia guidelines)					
empagliflozin	take	HOLD		HOLD	HOLD	HOLD
metformin	Take as usual					HOLD
Perindopril/indapamide	take	Take		take	Only take if >24h from OR	Hold
Insulin						Order some postop CBG / ss insulin
Rosuvastatin, tamsulosin	Take as usual					

POSTOP Mr. Uh-Oh – 71y diabetic no hx CAD/PAD, POD#2 ECG (radical cystectomy and L nephrectomy, non-cancer). Mildly nauseous, BP 110/60.



PREOP ECG



POSTOP ECG. hsTnI 20 → 7000 POD #2 – NO CHEST PAIN

Your hospital has no cardiologist, no cath lab.
What will you do?
Nearest cath lab is 120 min away.

Mr. Uh Oh Case summary

- Dx: Postop, relatively “asymptomatic” INFERIOR STEMI. Bleeding risk = very, very high.
- Transfer to ICU → Given ASA, IV heparin (after epidural catheter removed), Atorvastatin 80.
- Added clopidogrel once we were sure he wasn’t bleeding via ileal conduit on heparin infusion (POD#4).

TFR for cath : 70% distal LM and severe 3VD; underwent CABG x 4 ,14 days postop from his original nephrectomy.

Case of MINS screening, saving a life!

Steroids

- Not evidence based AT ALL
- Clinical gestalt / consensus
 - HPA Axis likely NOT suppressed:
 - Prednisone < 5 mg/d
 - Any dose < 3 wks
 - Uncertain
 - 5-20 mg prednisone > 3 wks (*some say > 5 mg for 3 wks = probably suppressed)
 - Consider ACTH stim test/consult Endo, or just give steroids as if suppressed (esp. if no time for testing and major surgical stress)
 - HPA Axis Suppressed: Prednisone $\geq$ 20 mg/d for 3+ weeks or Cushingoid
 - STRESS DOSE (this is “art of medicine” not evidence based – pick a number 50 or 100 and you will be fine in oral exam ;)
 - Major surgery – Usual AM dose + HC 100 mg IV X 1 pre-op, then 50 mg q8h X 3 doses, then 25 mg q8h X 3 doses then back to usual dose
 - Moderate surgery – Usual AM dose + HC 50 mg X 1 pre-op, 25 mg q8h X 3, then usual dose
 - Minor surgery – Usual AM dose

Pre-Op Blood Product Management

What if Hb was 95 and patient is a devout Jehovah's Witness who would not want transfusion.....

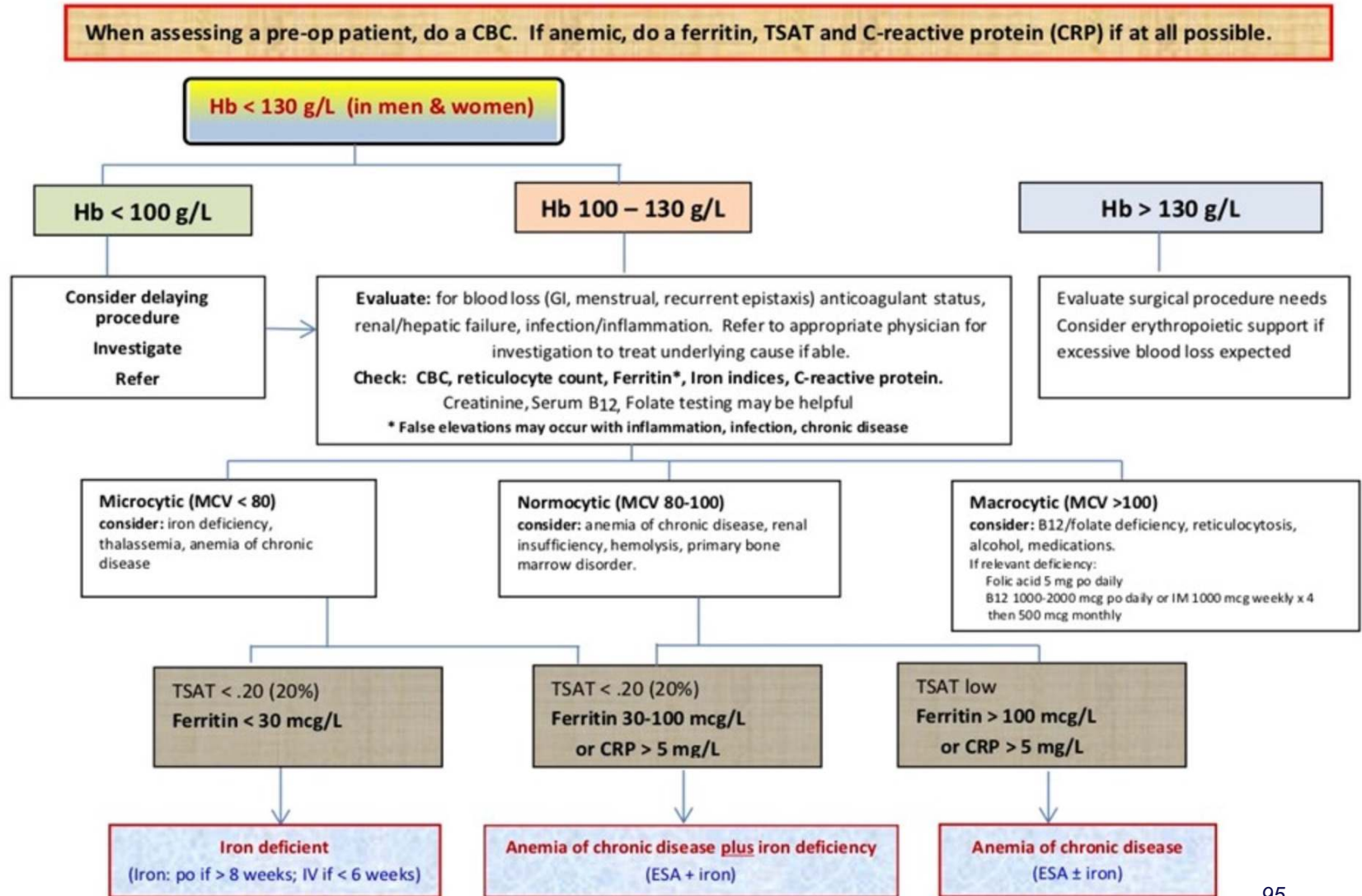
- Discuss patient preferences: involve surgeon, determine likelihood and magnitude of blood loss and discuss whether or not these risks can be mitigated surgically
- Clearly express possible consequences of NOT having transfusion and inquire if patient understands risks (ie. complications up to and including death)
- Document strategies and treatments available for this surgery (based on discussion with patient), delay surgery where possible to optimize anemia.
 - DECREASE PHLEBOTOMY (frequency, pediatric tubes)
 - Autologous blood donation pre-operatively [most pts with Jehovah's faith do not accept this practice, however you can discuss if available at your institution]
 - Tranexamic acid Pre-op and postop if bleeding complications or concerns re hemostasis
 - Acute Normovolemic Hemodilution (defer to anesthesia)
 - RBC Salvage
- Establish SDM. If you get "down to the wire" with life threatening bleed it is helpful to verify with SDM the patient really would NOT want blood in that scenario.

Anemia and Blood Product Management

“All patients undergoing [high bleeding risk] surgery OR DELIVERY should be screened for anemia at least 6 weeks prior to OR date”

- If there is a high bleeding risk associated with surgery (major Ortho, Gyne, GU, Vascular, CV surgery)
 - Preop CBC – add ferritin, Tsat, CRP if possible, consider Cr and B12
- If anemia (Hgb <130, in renal/cancer pts <120) is identified, treat as appropriate:
 - Erythropoietin stimulating agent +/- IV iron to reduce the need for allogeneic blood
 - Administer iron to patients with iron deficiency anemia if time permits
 - Note: IV iron still minimum 3-4 days for any effect

ONTRAC Ontario Blood Conservation Program



Decreasing Blood Loss and Transfusion

- Consider autologous transfusion
 - Patient may donate blood before admission only if there is adequate time for erythropoietic reconstitution
- If risk of excessive bleeding, consider tranexamic acid:
 - POISE 3 trial demonstrates TXA perioperatively reduces major bleeding risk
 - Meta-analyses of RCTs indicate that TxA for prophylaxis of excessive bleeding is effective in reducing perioperative blood loss, the number of patients transfused, and the volume of blood products transfused
 - RCTs comparing TxA with placebo or no TxA controls report no differences for stroke, myocardial infarction, VTE, renal failure, reoperation for bleeding, or mortality
 - Consider TxA prophylaxis for certain orthopedic procedures (such as knee replacement surgery), in liver surgery and other clinical circumstances at high risk for excessive bleeding (Postpartum Hemorrhage etc)

Decreasing Blood Loss and Transfusion

- Acute normovolemic hemodilution

[Typical procedure = After induction – remove a unit of blood and immediately replace w/ equal volume of fluid (usually 5% albumin), then at end of procedure auto-transfuse patient's blood back]

- Effective in reducing the volume of allogeneic blood transfused and the number of patients transfused for major cardiac, orthopedic, thoracic, or liver surgery
- Consider ANH **if available at your centre** in patients at high risk for excessive bleeding (*e.g.*, major cardiac, orthopedic, thoracic, or liver surgery)

- RBC Salvage / Cell Saver®

[Device that takes contents of intraop-suction, filters out tissue from blood, washes it, then recovered RBCs are re-transfused to pt]

- **Consider if available** and degree of anticipated blood loss justifies use
- Contraindications: site-specific (eg. bone cement), sepsis, malignancy

Patient Not Willing to Have RBC Transfusion

- Elicit patient's preferences clearly
 - May need privacy, patient alone
- Clearly express possible consequences of NOT having transfusion
- Describe other strategies and treatments available and what is acceptable to patient
 - DECREASE PHLEBOTOMY (frequency, pediatric tubes)
 - CHANGE THE SURGICAL PLAN (discuss with the surgeon)
 - Autologous blood donation pre-operatively
 - Tranexamic Acid, DDAVP
 - Acute Normovolemic Hemodilution
 - RBC Salvage
 - Iron, EPO, **optimize Hgb Pre-op before elective surgery**

FYI only : Perioperative BP Notes from POISE3 (2022)

BONUS

Read on own

POISE 3

– TXA 1g at start and end of surgery vs placebo

- OUTCOME: Composite of life threatening, major bleeding and bleeding into critical organ at 30d

= **Significant reduction of bleeding** (9.1% vs 11.7%, HR 0.76, $p < 0.001$)
[Devereaux PJ et al, NEJM 2022]. No difference in MINS.

- The Blood pressure arm (2x2 Factorial Design)

- MAP TARGET >60 or MAP TARGET >80 intraoperatively
- No difference in 30d CV outcomes

OSA and Peri-op Medicine

- Guidelines primarily address anesthetic techniques and airway management
- American Academy of Anesthesiologists suggestions:
 - Preanesthetic consultation in anyone suspected of sleep apnea.
 - Review of previous records, history and examination to determine likelihood of complicated airway management.
 - Anesthetist and surgeon should come to joint decision on whether to delay for sleep study (no specific criteria given on which to base decision - but consider in more invasive surgery and when post-op opiates needed)
 - In those at high-risk of OSA, consider inpatient rather than outpatient (ambulatory) OR. Especially if other cardiovascular comorbidities and/or need for post-op opiates.
 - If on CPAP, bring it in. If not responsive to CPAP, consider BiPAP.
 - If possible, perform regional/local anesthetic techniques to limit sedation and need for post-op opiates.
 - If possible, position patient in non-supine position during recovery.
 - Consider continuous pulse oximetry in those at risk of respiratory compromise.

Surgery In Special Populations (1)



- **Pregnancy**

- Emergent / Urgent non Obstetric surgeries (appendectomy, cholecystectomy, trauma etc.) occur frequently, anesthesia and OB consults (if fetal viability), IM usually not involved
 - Limited role for internist so not a likely scenario unless asked to do peri-op mgmt of another medical issue (VTE, Diabetes...).
- For semi-urgent surgery (ex breast cancer Lumpectomy) that cannot be delayed, general guidelines:
 - Try to do in 2nd trimester – no evidence that anesthesia causes birth defects or increases miscarriage risk but due to high rates of miscarriage in general population and surgeons not wanting to attribute preg loss to their procedure surgeries are delayed where possible to Tri2.
 - Third trimester – anatomic changes esp if upper abdo or head and neck surgery increase risk, + preterm labour risk. Continuous monitoring of all viable fetuses is generally done intraop.
 - Consider VTE prophylaxis for all pregnant patients postop
 - Be thoughtful about antibiotics used for surgical skin SSTI prophylaxis – eg no IV Vanco if PCN allergy, favour cephalosporin or clindamycin

Surgery in Special Populations (2)



- Cirrhosis

- COMPENSATED CIRRHOSIS: PREOP SCREEN for portal hypertension in compensated cirrhosis = liver stiffness and platelet count, imaging
- Degree of PORTAL HYPERTENSION and signs of decompensated cirrhosis (ascite, encephalopathy, SBP, varices) increase perioperative risk. Patients with decompensated cirrhosis have a high 1 year mortality rate in general, so elective surgeries are generally not offered
- Decompensated cirrhosis
 - Childs Pugh C, MELD >15 = elective surgery discouraged due to HIGH risk peri-op mortality/ morbidity.
 - Childs Pugh A, MELD 6-9, compensated cirrhosis “can undergo most surgical procedures with a reasonable safety profile with adequate pre-op optimization”
- Higher risk of mortality and morbidity for all types of surgery in patients w/ cirrhosis
 - E.g. Emergency Surgery mortality rate of 57% w/ decomp cirrhosis
 - Abdominal open procedures very high risk
- Considerations if undertaking surgery: Refer to liver centre where possible.
 - Bleeding risk – platelets <50-75 independent risk for bleeding (whereas INR does not correlate well to bleed risk). Ensure blood bank ready, avoid neuraxial anesthesia
 - VTE risk – high despite coagulopathy (elevated INR) and thrombocytopenia
 - Infection risk – higher at all surgical sites
 - Encephalopathy – common postop, esp w high doses of peri-op opioid analgesia or sedatives
 - Nutrition pre-op – high calorie (30-35 kcal/kg/d) and high protein (1.25-1.5 g/kg/d) at least 2 weeks preop
 - Frailty assessment preop
- If patient having abdominal or thoracic surgery with Clinically significant portal HTN consider preop TIPS procedure.
- If having bariatric surgery with compensated cirrhosis, SLEEVE GASTRECTOMY recommended.
- Control the usuals: strict ETOH abstinence, treat viral hepatitis, ensure varices screening up to date...

NEW in 2025 – ACG
Perioperative Guideline

Platelets <50 = give
thrombopoietin agonists
(ex eltrombopag) pre-op to
raise platelets.

Platelet count (not INR)
highly predictive of periop
bleeding risk.

Preop Vit K, blood products
for INR have not been
shown to reduce risk of
bleeding.

Surgery in Special Populations (3)



- Dialysis [stable ESRD, not acute emergency dialysis]
 - Most elective low risk procedures (e.g. AV Fistula) can proceed with following these general guidelines:
 - Ideally surgery day after routine dialysis (same day = heparin exposure, avoid Mondays for people on TTS schedule)
 - Potassium day of OR – for anesthesia considering paralytics - <5.5
 - Appropriate pre-op optimization of diabetes, blood thinners as in our previous slides. Caution if residual renal function to HOLD ACE/ARB/Diuretics peri-op to ensure no AKI with hypotension in procedure.
 - Nephrology managing inpatients postop who would usually be routinely under care of surgeon only (eg parathyroidectomy)
 - Urgent/Emergent procedures confer much higher risk of complication in ESRD population than those with normal renal function (ex. AAA open repair mortality 10X higher w/ ESRD)
 - Institute postop monitored bed, Establish Code Status / Goals of Care
 - Where possible achieve euvolemia, normal potassium/sodium pre-op

COVID & Perioperative Management

- Follow institutional / local public health guidelines on whether to use routine pre-op testing / surveillance for COVID-19 in pre-op patients
- Timing of elective surgery (ASA 2023 statement)
 - 2 weeks post infection if low risk surgery
 - High surgical risk or ongoing symptoms wait at least 7 weeks
- Patients with active COVID-19 undergoing surgery have poor outcomes
 - Anesthetic considerations for surgery: intubation is an AGMP

BONUS MCQ 1 2025

You are seeing a 67yM living with diabetes in pre-op consultation for advice regarding management of apixaban before an elective R TKA scheduled in 4 weeks time. On review of systems he has no cardiopulmonary symptoms. He has had no healthcare encounters recently except for a visit to the ER 2 weeks ago for lower urinary tract symptoms (nocturia, polyuria).

PMH – Obesity BMI>40, OSA, Hypertension, AFIB

Meds – Lisinopril, edoxaban 60 mg daily, ezetimibe 10mg daily.

148/92. On exam he has significant callous, onychomycosis and decreased sensation to monofilament in the feet.

Which of the following would be indications to delay elective surgery?

- a) You hear a Gr II SEM at the RUSB. Normal S1, S2, no radiation. Normal carotid exam.
- b) NT pro-BNP = 489 ng/L
- c) ECG showing inferior Q waves
- d) Glucosuria on recent urinalysis done when in the ER

BONUS MCQ 2, 2025

A 62yF with a bileaflet AVR for bicuspid aortic valve is referred for advice regarding warfarin before a colonoscopy. She had one 3 years ago showing high risk polyps that were snared.

She had a L central retinal artery occlusion in 2022 when INR subtherapeutic. She is in normal sinus rhythm with no history of AFIB. What do you suggest? BP 110/70, HR 60, sinus, wt 82kg Hgb 110 MCV 82, Plt 282

Which are the best recommendations to patient to follow for this outpatient procedure?

- a) Continue warfarin without interruption
- b) Hold warfarin 5 days pre-op, resume 12h post colonoscopy
- c) Hold 5 days pre-op. Take enoxaparin 80mg sc bid for 3 days before endoscopy, last dose morning before OR. Restart warfarin in evening post procedure, and LMWH anticoagulation postop day 1
- d) Hold 5 days pre-op. Take enoxaparin 80 mg and warfarin 12h post colonoscopy.

BONUS MCQ 3 - 2025

A 76yF is seen prior to elective hip arthroplasty. She had a previous inferior STEMI w TNK and RCA drug eluting stent in 2021. She is on prasugrel ramipril, rosuvastatin, vitamin D. She is a non-smoker. She gets short of breath on exertion, stable for years. She undergoes a stress test where she completes Stage 1 of Bruce Protocol to target heart rate but it shows 1mm ST depression in II, III, AVF. Your cardiac exam is normal. BP 128/88, HR 87. What do you recommend with regards to her perioperative management:

- a) No further testing or changes
- b) Rivaroxaban 2.5mg po bid for perioperative vascular optimization
- c) Referral to cardiology for PCI prior to surgery
- d) Start and titrate beta-blocker, proceed to OR
- e) Hold prasugrel 5 days prior to surgery

BONUS MCQ 4 2024

You see an obese pregnant 26yF for gestational diabetes. You detect a thyroid nodule on your exam. Ultrasound reveals a 1.6 cm hypoechoic left thyroid nodule without calcifications. No adenopathy or family history of thyroid cancer. EXAM: BP 110/70, HR 90, wt 110 kg.

Meds: ASA 162mg HS for preeclampsia prevention, metformin 1 g BID, prenatal vitamins, ferrous sulphate 300 mg daily, insulin NPH @ bedtime

FNA reveals a focus of follicular thyroid cancer. ENT recommends a hemithyroidectomy. She is now 35 weeks GA. Which of the following would you recommend?

- a) HOLD ASA for 72 h prior to surgery
- b) Postop Pharmacologic DVT prophylaxis with dalteparin 5000 u sc BID to start POD#1
- c) Delay surgery until postpartum.
- d) Consult OB for recommendations regarding intraoperative fetal monitoring

MCQ #5 (2024)

You are seeing a 78y patient who broke their hip today and they are going to the OR tomorrow evening. They have a history of stable coronary artery disease with 3 stents 10 years ago and remote TIA. Diabetes on oral agents. Known L carotid stenosis 60%. Hypertensive. He is on azathioprine for severe mixed connective tissue disease with arthritis. He is in pain. BP 108/72, HR 62.

Medication reconciliation is listed here:

ASA 81 mg daily, clopidogrel 75 mg daily, atorvastatin 80 mg daily, ramipril 5mg daily, metoprolol 25mg bid, metformin 1g bid, azathioprine 75mg daily.

With regards to admission medication reconciliation, which of the following is incorrect:

- a) Hold antiplatelets
- b) Hold ramipril
- c) Hold metformin
- d) Hold azathioprine
- e) Increase metoprolol to 100 mg BID

MCQ #6 2024

A 72y M is referred for preop assessment. The surgeon wonders if they can stop clopidogrel prior to upcoming elective laparoscopic cholecystectomy for symptomatic gallstones.

PMH: Lacunar stroke 2 years ago. HTN, Dyslipidemia, ex-smoker. No history of cardiac disease or Afib. Osteoarthritis. Last known LVEF 65%.

Meds: Clopidogrel 75mg daily, Indapamide 2.5mg daily, Simvastatin 40 mg daily, vitamin D, Ibuprofen PRN arthritis.

On exam: Alert, oriented. Using his iPhone he shows you his BP and HR trend for past 2 years which he tracks meticulously. BP 128/72, HR 68. Cardiac exam normal.

ECG shows normal sinus rhythm with ECG LVH.

With regards to his pre-operative management, which is the next best recommendation?

- a) Advise to hold clopidogrel 5 days preoperatively
- b) Measure NT-pro BNP
- c) Order post-operative ECG and Troponin
- d) Order exercise stress test

BONUS MCQ #7 2023

A 63yF is scheduled for an elective TKA for severe Rt knee Osteoarthritis in 6 days. She has a history of stable CAD following NSTEMI and PCI with 1 stent in 2018. She reports she is short of breath when she walks up the stairs but this has been present for over one year as she has significant weight gain due to lack of exercise with painful knee arthritis.

BP 110/68, BMI 37, HR 58. CV Exam unremarkable.

Meds: Clopidogrel 75 mg daily, PRavastatin 10mg daily, Bisoprolol 5mg daily, Lisinopril 10 mg daily, celecoxib 100 mg po BID, acetaminophen 1 g po TID.

ECG: Sinus Brady, old inferior Q's.

Labs: Hgb 109, Cr 83, A1C 6.3% , NT pro BNP 120 mg/L

With regards to preventing myocardial injury after surgery, what is the best recommendation at this time?

- a) Advise anesthesiologist to target MAP >80 intraoperatively
- b) Advise anesthesiologist to deliver FiO2 >= 80% intraoperatively
- c) Advise anesthesiologist to keep temperature 37oC intraoperatively
- d) Delay surgery
- e) Continue bisoprolol perioperatively

Bonus MCQ #8-2025

With regards to postop thromboprophylaxis, which is not correct?

- a) Postop Transurethral resection of the prostate, catheter in situ, heparin 5000 u sc BID
- b) Postop elective THA, rivaroxaban 10mg x 5 d then ASA 81mg x 30d
- c) Postop total femoral nail, heparin 5000 u sc BID
- c) Postop hemicolectomy for stage III colorectal CA, dalteparin 5000 u sc daily

MCQ #9 2025

A 78yF is referred for assessment prior to a R hemicolectomy for early stage colon CA. She was treated in 2023 for a NSTEMI with PCI to LAD x 1. She reports no CV symptoms, colon CA detected when she was diagnosed with iron deficiency anemia by family doctor. She had dyspnea on exertion which improved with oral iron.

Meds: Clopidogrel 75mg daily, rosuvastatin 40mg daily, Bisoprolol 5mg daily, Ramipril 10mg daily, lorazepam 1mg HS prn, ferrous fumarate 300 mg OD
152/98, HR 78, 99% RA. Cardiac exam is normal. BMI 28.

Labs: Hgb 108 Plt 400 ferritin 10, NT pro BNP 245 ng/L, Cr 78

ECG Normal sinus rhythm

Which of the following is next best step?

- a) Hold ramipril perioperatively
- b) Give IV iron prior to surgery
- c) Order post-operative Troponin and ECG x 48-72h
- d) Order exercise stress test urgently prior to OR

MCQ#10 2026

A 65yM man presents 2 weeks prior to a transurethral resection of prostate for benign prostatic hypertrophy. Review of his chart demonstrates a recent embolic R ACA stroke consequent to right internal carotid stenosis (68%, long segment of stenosis from carotid bifurcation to intracranial vessels, not surgical candidate, not EVT/stenting candidate) 4 weeks ago. He received IV thrombolysis for acute stroke and has minimal residual deficits.

Current medications include clopidogrel 75mg daily, simvastatin 40mg daily, candesartan/hctz 32mg/25mg daily, tamsulosin 0.4mg daily. BP is 146/92, HR 58 bpm.

Which of the following is most appropriate?

- a) Change clopidogrel to ASA 81mg perioperatively
- b) Hold clopidogrel 5 days prior to planned surgery
- c) Defer surgery at least 3 months in view of recent stroke
- d) Add amlodipine 5mg daily to optimize BP perioperatively

MCQ #11 2025

A 65yM is referred for pre-op evaluation before an urgent cystectomy for high risk muscle invasive bladder cancer. He had a NSTEMI 2 months ago and has 2 drug eluting stents. He has been having gross hematuria passing clots since starting cardiac meds below. Denies chest pain or dyspnea, able to rake leaves over weekend.

Meds: ASA, ticagrelor 90mg BID, simvastatin 40mg OD, Ramipril 10mg OD, Tamsulosin 0.4mg daily, Nitro pRN (Not using)

BP 108/76, HR 62 regular, BMI 24, Cardiac exam normal.

Please choose best recommendation below regarding perioperative cardiovascular optimization.

- a) Delay surgery 30 days until 3 months post PCI
- b) Hold ticagrelor x 5d, hold ramipril 24h, continue ASA
- c) Obtain urgent pharmacologic stress test to risk stratify
- d) Continue dual antiplatelets, group and crossmatch x 2 units, notify anesthesia to avoid neuraxial anesthesia

BONUS MCQs Answers

- 1) **(D)** – this patient likely has diabetes and it may be decompensated based upon physical exam and polyuria with glucosuria. Consider A1C and delay to optimize.

While exam shows a murmur it does not have features of severe AS. Surgery in 4 weeks so you can likely get an ECHO if you feel it might change management.

Q waves and elevated NT pro-BNP suggest higher risk of perioperative MACE, but these are not indications to delay (rather to counsel, optimize risk factors, plan postop monitoring and so forth)

- 2) **(C)** – this pt has high risk as has had previous embolic stroke when INR subtherapeutic (implied by stem because she was on antibiotics when this happened). Colonoscopy on its own is not high risk but it's suggested she will need Polypectomy (previously done 3y ago, microcytic anemia...) and this is a high risk for bleeding procedure so you do need to interrupt anticoagulation and Recommend bridging.

The tough part is when giving patients concrete advice you have to make recommendations not knowing what will happen in surgery – hence assume she will have polypectomy, hold warfarin, bridge w LMWH preop, can restart VKA 12h post procedure but delay LMWH til next day as polypectomy high risk (ideally endoscopist will advise in truth in recovery room, we hope!).

- 3) **Pick D** - STABLE CAD. The bigger question is why did she get a preop stress test with stable symptoms? Rivaroxaban 2.5 BID per COMPASS trial not indicated to start periop. Stable CAD no role for PCI. Prasugrel needs to be held, 7-10 days for spinal, change to ASA 81mg. How long to wait for OR? Depends on clinical scenario. If patient is booked for surgery tomorrow you really don't have time to start – but if OR in a couple weeks can start today eg bisoprolol 2.5 daily, reassess in a week ...
- 4) **PICK C**. This woman is 35 weeks pregnant. Due to anatomic changes in pregnancy head and neck surgery at this time is VERY DANGEROUS. This is a very contrived scenario as no ENT in their right mind would book her for procedure right now. She will be delivered in 3-4 weeks given she has GDM and can have surgery delayed postpartum. This is a low risk (eg not anaplastic) thyroid cancer discovered incidentally in pregnancy and the cancer is not nearly as life threatening as the surgery would be if done at this gestational age. With regards to her ASA she is on this for preeclampsia prevention so will stop it at 36 weeks so this doesn't apply to preop scenario

BONUS MCQ Answers

5) Answer: E The patient is going for high risk bleeding surgery tomorrow and is already "on" DAPT. You can reassess antiplatelets postop. Anesthesia should be aware of clopidogrel exposure = No epidural. The pt's HR and BP are low so per CCS recommendations and knowing results of POISE high dose metoprolol not wise

6) Answer A: This is low risk OUTPATIENT laparoscopic surgery. Does not need BNP.
Remote stroke. No other indication for antiplatelet.

7) Delay surgery - This patient has significant symptomatic anemia and is scheduled for surgery with high risk of major bleeding (>1L average blood loss). At this time safest to delay surgery to optimize Hgb – major anemia puts pt at high risk of "type 2" supply/demand mismatch MINS or even ACS *ALSO* - clopidogrel on board, ideally for anesthesia to give spinal want held 7 days There are RCTs which show (a) and (b) and c don't reduce MINS. Bisoprolol can be continued perioperatively though "at this time" you don't want to greenlight elective surgery in a cardiac patient who is not optimized.

8) TURP – shouldn't require prophylaxis per guideline.. Catheter in situ suggests there was some bleeding complication potentially. TFN = hip fracture nail, so while LMWH is better we didn't get Creatinine so LDUH is also not wrong.

9) B - Symptomatic Iron Deficiency Anemia requiring urgent correction before this surgery. NTproBNP is below cutoff of concern (300 ng/L) so per guidelines one does not have to do routine MINS screening postop. We will talk about new data re: holding RASi argue to hold but BEST thing to do is fix IDA.

10 C - C - With recent stroke and encountering purely elective surgery it may be prudent to delay and during this time optimize further. As the underlying cause for stroke (carotid stenosis) cannot be fixed this puts him at higher risk.

Note that amlodipine and SIMVAstatin interact – amlodipine inhibits CYP3A4 which can result in higher SIMVA levels in blood.

11) B - In real life and on exam: Discuss with their cardiologist, notify anesthesia, discuss with surgeon ... plus optimize all the other risk factors. Discuss w anesthesia ticagrelor 5 vs 7 days – cystectomy often offered neuraxial anesthesia